# When RGB Meets IR: Orthogonal Disentanglement and Dual-Expert Enhancement for Day-to-Night Aerial Object Detection

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***Abstract*—Day-to-night RGB–infrared (IR) fusion object detection on UAV platforms is a crucial approach for overcoming the limitations imposed by extreme illumination conditions and complex environments, enabling reliable environmental perception at all times. This capability plays a vital role in advancing the practical deployment of UAVs in applications such as security surveillance, emergency rescue, and autonomous navigation. However, existing methods commonly suffer from cross-modal geometric misalignment and rely on simplistic illumination-based adaptive fusion strategies, which fail to fully exploit the complementary characteristics of different modalities. To address these challenges, we propose Disentanglement$^2$ Enhancement$^2$-Detector (D$^2$E$^2$-Det), which incorporates two novel modules: a cross-modal orthogonal disentanglement (COD) module for separating shared and modality-private features to enable precise cross-modal alignment and a dual-expert complementary enhancement (DCE) module for enhancing modality-specific discriminative features through an illumination-aware gating mechanism. Extensive experiments conducted on three benchmark datasets—DroneVehicle, RGBTDronePerson, and VTUAV-det—demonstrate that D$^2$E$^2$-Det achieves mAP scores of 61.10%, 19.42%, and 35.09%, respectively. These results significantly outperform existing state-of-the-art methods, validating the effectiveness of the proposed approach for robust object detection in day-to-night and complex UAV scenarios. The code is available at https://github.com/HL202604/D2E2-Det**

***Index Terms*—Aerial images object detection, lighting perception, multimodal fusion, object alignment.**

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## I. Introduction

AERIAL object detection [1], [2] serves as a key perception technology for UAV autonomous navigation [3], intelligent security [4], disaster response [5], and aerial search and rescue [6], [7] and holds significant practical value in real-world applications. However, UAV-based remote sensing platforms often encounter complex environmental conditions, such as day-to-night variations and fog or haze [8], [9], during Earth observation missions. Under these circumstances, the limitations of single-modality remote sensing data become increasingly evident: the signal-to-noise ratio of RGB imagery deteriorates significantly under low-illumination conditions, while infrared (IR) data are prone to thermal diffusion and cross-interference in daytime scenes, resulting in blurred target representations. Consequently, developing robust [10] detection algorithms that can adaptively fuse multimodal remote sensing information is essential for improving the accuracy and reliability of day-to-night remote sensing monitoring.

To address these challenges, RGB–IR fusion-based detection methods have gradually developed along two core trajectories: cross-modal feature alignment and illumination-aware fusion. In feature alignment, early research focused on solving physical spatial offset problems through explicit geometric modeling. For instance, the TSRA module [11] learns transformation parameters at the region proposal level to correct geometric biases, while the DPAL method [12] introduces multiscale dilation offsets to capture misalignment at different scales. More recent research has shifted toward more efficient, global alignment mechanisms. For example, C$^2$Former [13] employs cross-modal attention for semantic-level soft alignment, while CLGNet [14] utilizes dedicated calibrated convolutional layers for end-to-end spatial position adjustment, significantly improving computational efficiency without sacrificing accuracy. In adaptive fusion, early work established illumination awareness (IA) as the core guiding principle. The IGT framework [15] generates continuous fusion weights through an illumination-aware subnetwork, while RISNet [16] enhances modality complementarity by minimizing mutual information. Subsequent studies have further refined the fusion process. TINet [17] designs Inter-MA and Intra-MA modules to extract enhanced features across and within modalities, respectively, while LAIIFusion [18]

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achieves fine-grained fusion control at the grid level through a local adaptive mechanism, significantly enhancing the model's adaptability in complex scenarios. These systematic explorations have laid a solid foundation for robust day-to-night detection tasks.

While existing studies establish a solid foundation for RGB–IR day-to-night object detection, two critical bottlenecks still hinder the efficient fusion of complementary modal information under extreme imaging conditions, as illustrated in Fig. 1(a) and (b). These bottlenecks correspond to the following challenges.

1) *Degradation of Modality-Private Features During Alignment:* Most existing cross-modal alignment methods operate directly on entangled features that contain both shared geometric information and modality-private characteristics. As a result, the estimation of geometric transformations becomes vulnerable to interference from modality-dependent factors, such as RGB texture details and IR thermal diffusion artifacts, which hinders accurate pixel-level spatial alignment.
2) *Insufficient Exploitation of Modal Complementarity:* Although mainstream fusion strategies incorporate illumination-aware mechanisms, they essentially remain at the level of reweighting original modal features. This selective weighting paradigm fails to explicitly mine and enhance modality-private discriminative features under varying environmental conditions, thereby limiting the performance upper bound of existing models in complex day-to-night scenarios.

In this article, we propose a novel method for day-to-night RGB–IR object detection in UAV scenarios, termed Disentanglement$^2$ Enhancement$^2$-Detector (D$^2$E$^2$-Det), as illustrated in Fig. 1(c). The proposed framework addresses two fundamental challenges in RGB–IR fusion: geometric noise interference during cross-modal alignment and insufficient exploitation of modality-private discriminative power in existing fusion strategies. These issues are mitigated through two innovative modules. First, we design a cross-modal orthogonal disentanglement (COD) module that explicitly disentangles modality-shared features from modality-private features via orthogonality constraints. This design enables accurate alignment based on clean geometric structures while fundamentally suppressing interference from modality-private noise. Second, we develop a dual-expert complementary enhancement (DCE) module, which dynamically selects and enhances the discriminative outputs of RGB and IR experts through an illumination-aware gating network. This mechanism enables a paradigm shift from conventional weighted selection to explicit complementary enhancement. Extensive quantitative and qualitative experiments on three publicly available RGB–IR benchmark datasets demonstrate that D$^2$E$^2$-Det achieves state-of-the-art performance in terms of both detection accuracy and robustness under challenging day-to-night conditions. The main contributions of this article are summarized as follows.

1) We propose a novel RGB–IR day-to-night object detection method, termed D$^2$E$^2$-Det, which improves detection stability under complex environmental conditions through a collaborative design of decoupling-before-alignment and dual-expert enhancement.
2) We systematically develop a decoupling and enhancement-oriented methodological framework that consists of two key components: a cross-modal orthogonal decoupling module that achieves high-purity separation of shared and modality-private features via a cosine-similarity-guided complementary gating mechanism, and a DCE module that selectively strengthens modality-private discriminative representations through an illumination-aware gating network and dedicated expert branches.
3) We further analyze the adaptability of the proposed modules across different detection paradigms. Results show that D$^2$E$^2$-Det consistently improves CNN-based one- and two-stage detectors, while its effectiveness on Transformer-based detectors is relatively limited.

## II. Related Work

### A. Drone-Based Aerial Object Detection

Drone-based aerial object detection has attracted widespread attention in recent years and, owing to its flexibility and ability to efficiently cover large areas, is widely applied in fields such as traffic monitoring, agricultural surveying, and edge-deployed sensing [19]. Traditional aerial object detection methods [20], [21], [22] often rely on single-modal image data; however, factors like lighting variations and weather conditions frequently interfere, making the RGB modality insufficient for detection in extreme scenarios, such as nighttime or hazy weather, thereby limiting its practical application. Multispectral image fusion techniques, combining RGB and IR, have achieved success in pedestrian detection [23], [24], [25], [26]. However, the scarcity of data has hindered progress in utilizing multimodal information for aerial object detection. Recently, the continuous emergence of datasets such as VEDAI [27] and DroneVehicle [28] has accelerated research into multimodal fusion techniques, addressing more challenging aerial object detection tasks involving unique perspectives and complex backgrounds. This has become a rapidly developing research hotspot.

### B. Cross-Modal Object Alignment

Cross-modal object alignment [29] aims to address geometric misalignment between RGB and IR images caused by differences in sensor characteristics, viewing perspectives, and temporal acquisition [30] and serves as a prerequisite for effective multimodal feature fusion. In the past two years, numerous approaches have been proposed to alleviate the performance degradation induced by misalignment. The DSAB module [31] reduces interchannel discrepancies and enriches unimodal feature details by explicitly designing bidirectional multichannel semantic alignment blocks. GAANet [32] introduces an IAM module to perform secondary calibration and local region comparison after graph-based node interaction, enabling accurate cross-modal target perception and alignment while improving detection accuracy with fewer

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parameters. The AMDANet architecture [33] adopts low-rank weighted features as alignment cues and employs a semantically consistent inference mechanism to reduce intermodal discrepancies, demonstrating strong performance in semantic segmentation tasks. MuIFS-CAP [34] constructs a correlation matrix during the cross-modal object alignment stage to model pixel-level associations across modalities, thereby improving the effectiveness of cross-modal fusion. The CIFDet algorithm [35] incorporates a dual-branch correlation-guided information fusion module, where a global correlation branch searches the entire feature map to handle severe geometric misalignment, and a local perception branch focuses on subtle scale variations. This design achieves robust detection performance on both public benchmarks and self-collected datasets.

Overall, although existing studies advance cross-modal alignment techniques, they commonly optimize within an undifferentiated mixed feature space. This practice introduces a fundamental contradiction in alignment networks: while correcting geometric offsets, the models must simultaneously contend with noise arising from modality-private information, such as texture details in RGB images and thermal artifacts in IR imagery. As a result, geometric alignment is often achieved at the expense of blurring or weakening modality-private discriminative features that are critical for accurate recognition. Consequently, effectively isolating noise interference while preserving the discriminative capacity of each modality during precise spatial alignment remains a key and unresolved challenge.

### C. Illumination-Aware Cross-Modal Fusion

Cross-modal adaptive fusion aims to dynamically integrate complementary information from RGB and IR modalities according to imaging conditions and semantic context and constitutes a core component for achieving robust day-to-night object detection. In recent years, a large body of research has centered on the principle of IA in fusion design, which enhances both the physical interpretability and effectiveness of multimodal detection models. EMOD [36] improves RGB–IR perception accuracy by implicitly embedding binary illumination cues into the fusion module to guide attention computation, and this strategy can be flexibly integrated into various advanced detection frameworks. To alleviate local detail degradation caused by inaccurate illumination guidance, FIAFusion [37] incorporates a feedback subnetwork that feeds illumination information back into the fusion module for illumination correction, thereby optimizing the fusion process and outperforming state-of-the-art multimodal pedestrian detection methods. Xu et al. [38] propose a multimodal sensor fusion approach that embeds an illumination-aware neural subnetwork to dynamically adjust fusion weights based on illumination cues, enabling reliable day-to-night detection. CAFN-IA [39] integrates a light-sensing module that combines quantified illumination intensity with the IoU metric across modalities to dynamically adjust predicted bounding box positions, effectively supporting UAV-based detection tasks. Additionally, to adapt detection models to agricultural pest monitoring under extreme weather and complex lighting conditions, another study [40] develops an environment-guided RGB–IR channel selection and weighting mechanism, further demonstrating the importance of illumination-aware fusion in challenging real-world scenarios [41], [42].

However, although these methods incorporate IA, their core design remains confined to a selective fusion paradigm that primarily reweights raw or shallow features. Such approaches mainly address the question of when to emphasize a particular modality while neglecting the more critical issue of how to effectively exploit the selected modality. As a result, the discriminative potential of each modality under specific environmental conditions is not fully activated, which constrains the performance upper bound of existing models in complex scenes. Consequently, developing a fusion mechanism that goes beyond static or adaptive weighting and explicitly enhances modality-private information emerges as a key factor for improving detection robustness in day-to-night scenarios.

## III. Proposed Method

### A. Overall Architecture of $D^2E^2$-Det

The overall architecture of the proposed $D^2E^2$-Det method is shown in Fig. 2. This network employs a dual-stream input design to process RGB and IR modal input images in parallel. In the feature extraction stage, $D^2E^2$-Det introduces a COD module that performs modal alignment on the RGB and IR feature maps at three different downsampling stages. The COD module explicitly aligns and separates shared and unique features between modalities through orthogonal decoupling, effectively alleviating the spatial misalignment problem caused by differences in sensor object positions and imaging principles. This approach lays a solid and high-quality feature foundation for subsequent multimodal information fusion.

In the neck network of the feature pyramid, $D^2E^2$-Det utilizes DCE modules on the three feature maps with 256, 512, and 1024 channels to fully leverage the advantages of each modality. The DCE module uses the illumination evaluation result of the original RGB image as a switch to select the appropriate expert and utilizes the precisely aligned features of the RGB and IR streams, processed by the corresponding COD modules in the backbone network, to enhance the main modal features during both day and night.

The $D^2E^2$-Det method achieves progressive multimodal fusion from low to high levels, leveraging strengths and mitigating weaknesses. It fully utilizes the information from both RGB and IR modes, creating a $1 + 1 > 2$ effect and ensuring accurate detection of various objects under day-to-night aerial photography perspectives.

### B. COD Module

In multimodal object detection of aerial images, there is a significant spatial misalignment between RGB and IR images, generally involving three dimensions: angle, size, and position, as shown in Fig. 3(a) and (d). This is mainly attributed to the fundamental differences in spectral response characteristics and imaging mechanisms between RGB and IR sensors, as well as the unavoidable slight differences in installation position during actual deployment. Direct cross-modal feature fusion introduces noise interference and reduces detection

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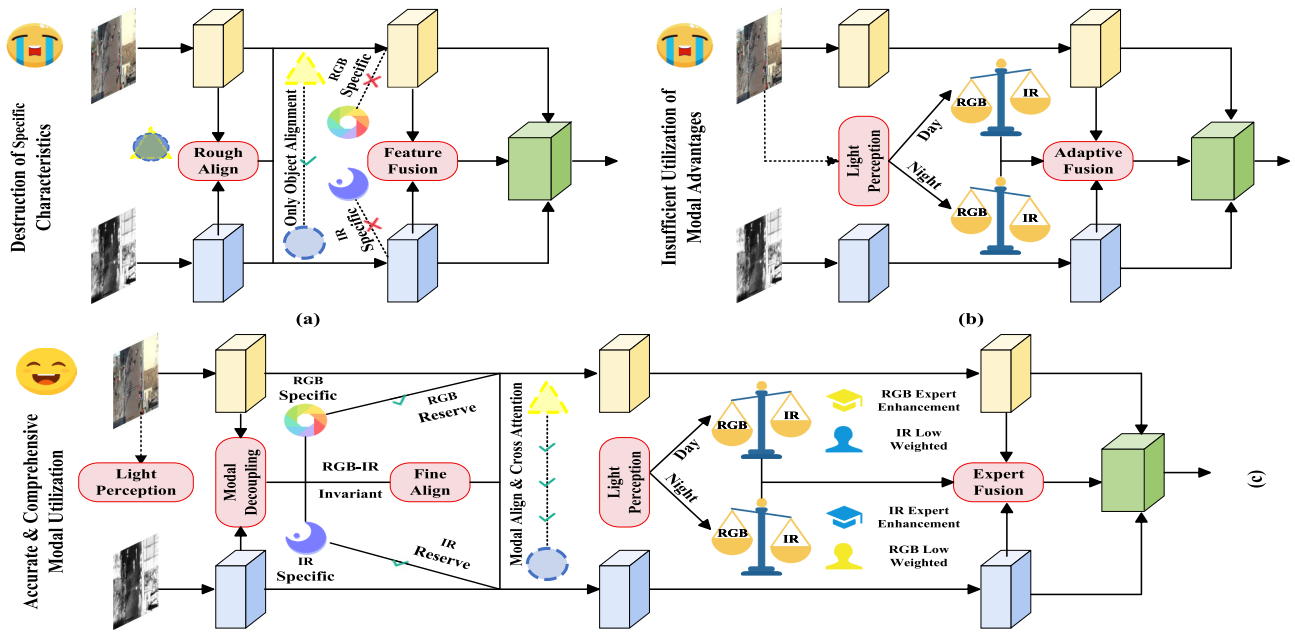



Fig. 1. Differences between the D$^2$E$^2$-Det method proposed in this article and the two mainstream RGB and IR multimodal fusion methods. (a) Modality-aligned fusion. (b) Illumination-aware fusion. (c) D$^2$E$^2$-Det (ours).

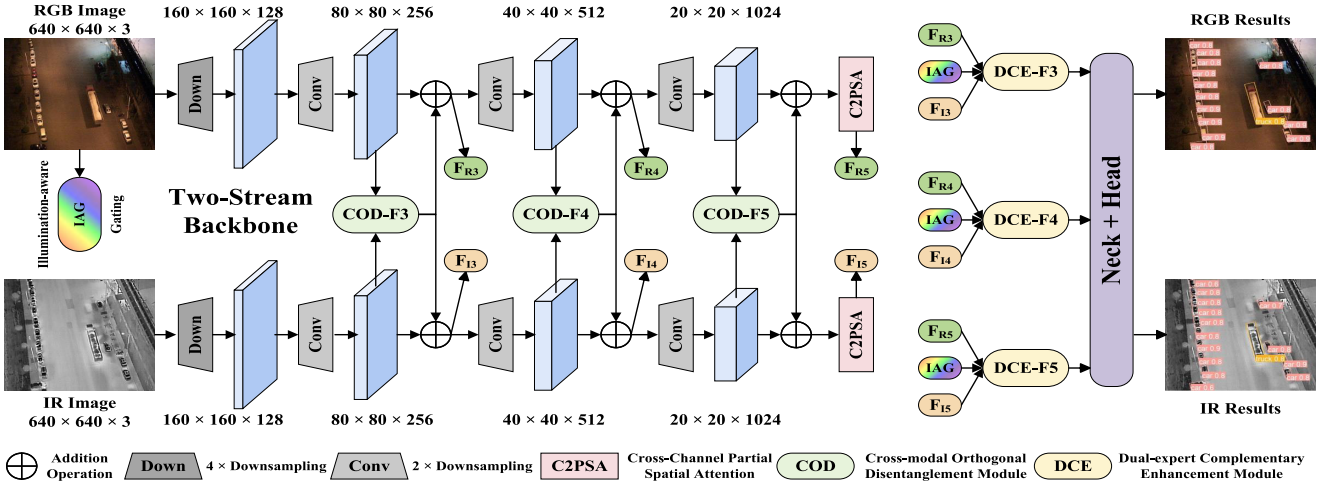



Fig. 2. Overall architecture of our D$^2$E$^2$-Det method mainly integrates COD for cross-modal feature decoupling and precise alignment in the two-stream feature extraction stage, and DCE for illumination-aware guidance is embedded in the neck interaction stage.

accuracy. Traditional solutions such as deformable convolution (DCN) focus on local geometric deformation alignment, while cross attention (CA) emphasizes the soft alignment of semantic information. However, neither explicitly reveals the intrinsic separation between modality-shared information and modality-private information at the feature representation level. To address this limitation, we propose the COD module shown in Fig. 4, which innovatively decomposes cross-modal features into modality-shared and modality-private components and formulates the ideal decoupling objective of these two components in an orthogonal manner

$$\mathcal{F}_m = \mathcal{F}_m^s \oplus \mathcal{F}_m^p \quad \text{s.t.} \quad \left\langle \mathcal{F}_m^s, \mathcal{F}_m^p \right\rangle = 0 \tag{1}$$

where $\mathcal{F}_m \in \mathbb{R}^{C\times H\times W}$ denotes the feature of modality $m \in \{\text{RGB}, \text{IR}\}$; $\mathcal{F}_m^s$ and $\mathcal{F}_m^p$ denote the shared and private features, respectively; and $\oplus$ represents the straight-line decomposition. This decomposition forces the network to learn more discriminative feature representations. Shared features encode cross-modal consistent information (such as object contours and spatial locations), while private features capture modality-private information (such as RGB texture colors and IR thermal radiation distribution). In implementation, the COD module does not introduce an additional explicit orthogonality loss. Instead, it employs a complementary gating mechanism to softly separate shared and private information, thereby approximately achieving the decoupling objective described above.

*1) Orthogonal Decoupling Based on Cosine Similarity:* To achieve the above decomposition, we introduce a position-encoded enhanced cosine similarity metric. First, we map the input features to initial shared and private candidate features using a learnable projection

$$H_m^s = W_m^s \mathcal{F}_m, \quad H_p^m = W_m^p \mathcal{F}_m \tag{2}$$

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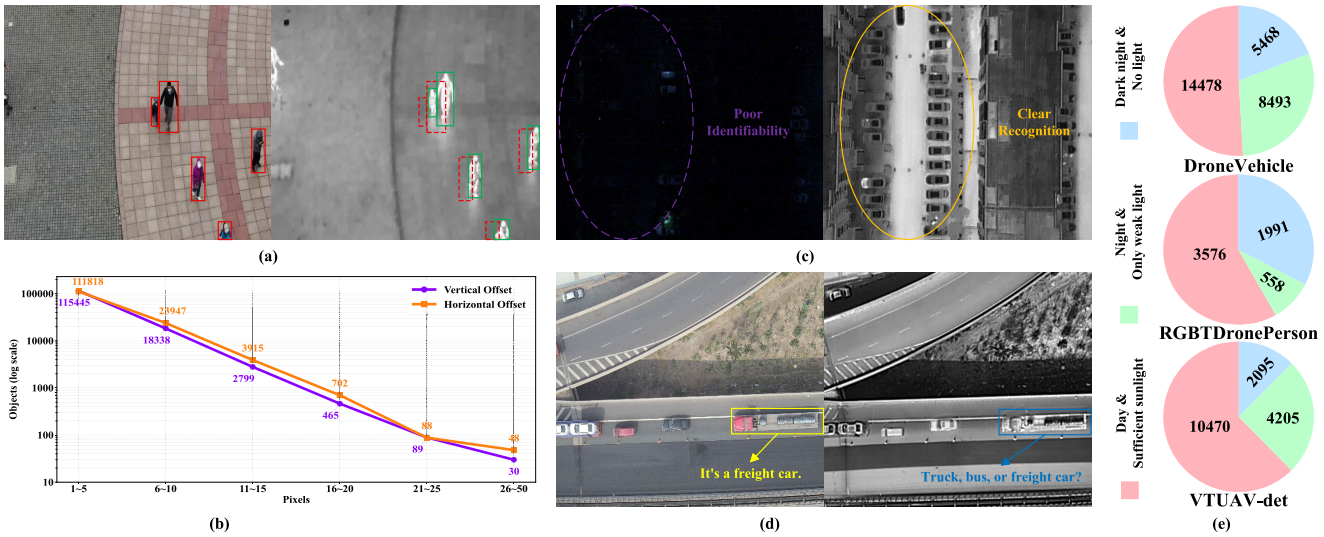



Fig. 3. Two key challenges in RGB-IR multimodal object detection in aerial images: 1) how to eliminate the misalignment of feature angles, positions, and sizes between the two modalities in (a) and (d) due to sampling equipment limitations before cross-modal fusion and 2) how to effectively utilize the advantageous information from the nighttime IR modality in (b) and the daytime RGB modality in (c) to guide cross-modal fusion. (a) Spatial misalignment of cross-modal objects. (b) DroneVehicle dataset object offset statistics. (c) Advantages of IR modal in dark night situations. (d) Advantages of RGB modal in day situations. (e) Temporal image statistics.

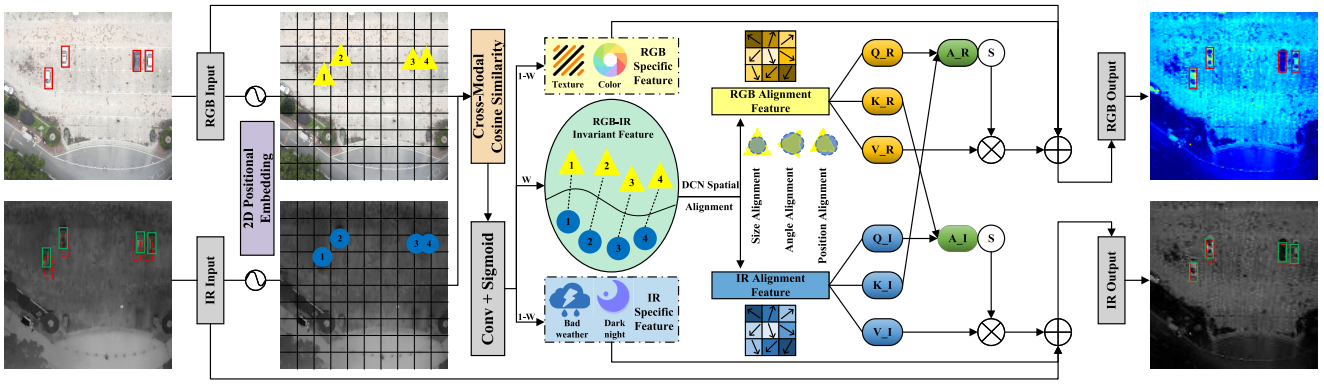



Fig. 4. Detailed process of the COD module distinguishes between shared and private features through orthogonal decoupling, aligns shared features, and retains private features, laying the foundation for subsequent integration.

where $W_m^s, W_m^p \in \mathbb{R}^{C\times C}$ represents the learnable projection weights and $\mathcal{P} \in \mathbb{R}^{D_p\times H\times W}$ is a 2-D sinusoidal positional encoding with a channel dimension $D_p$ preset to a fixed value (64 in this article). This encoding is then concatenated and fused with the initial shared features after adjusting the number of channels through linear projection

$$\tilde{H}_m^s = H_m^s + W_p\mathcal{P}. \tag{3}$$

This step ensures that the shared feature $\tilde{H}_m^s$ already contains spatial location information in subsequent calculations. Then, based on the location-enhanced shared features, we calculate the normalized cross-correlation matrix between the RGB and IR modes as a cross-modal similarity measure

$$S_{ij} = \frac{\left\langle \tilde{H}_{\text{rgb}}^s(i), \tilde{H}_{\text{ir}}^s(j)\right\rangle}{\left\|\tilde{H}_{\text{rgb}}^s(i)\right\| \left\|\tilde{H}_{\text{ir}}^s(j)\right\|} \tag{4}$$

where $i$ and $j$ represent spatial locations and $\langle\cdot,\cdot\rangle$ represents the vector inner product. The similarity graph $S \in \mathbb{R}^{HW\times HW}$ reflects the spatial correspondence of cross-modal features. Based on this, a gated generative network $\mathcal{G}$ is designed, which outputs a spatially adaptive gated weight graph $G_m \in [0,1]^{C\times H\times W}$, taking the global statistics of the similarity graph as its output

$$G_m = \sigma\left(\mathcal{G}\left(\text{Pool}\left(S\right)\right)\right) \tag{5}$$

$$\mathcal{F}_m^s = G_m \odot \tilde{H}_m^s, \quad \mathcal{F}_m^p = (1 - G_m) \odot H_m^p \tag{6}$$

where $\sigma$ is the sigmoid function and $\odot$ denotes the element-wise multiplication. This mechanism ensures that the shared feature $\mathcal{F}_m^s$ focuses on spatially aligned cross-modal information, while the private feature $\mathcal{F}_m^p$ retains modality specificity.

*2) Spatial Alignment Guided by Cross-Correlation:* Even after decoupling, the shared features $\mathcal{F}_{\text{rgb}}^s$ and $\mathcal{F}_{\text{ir}}^s$ still exhibit spatial offset. Therefore, COD utilizes the cross-correlation matrix $S$ to guide DCN for precise alignment. First, global alignment cues are extracted from $S$

$$g = \text{GlobalPool}\left(S\right) \tag{7}$$

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where $g$ denotes the global alignment cue obtained from the cross-correlation matrix. The alignment cue is then fused with the shared features and input into the offset prediction network

$$\Delta = f_{\text{offset}}\left(\left[\mathcal{F}^{s}_{\text{rgb}} \cdot \mathcal{F}^{s}_{\text{ir}}; \text{Repeat}\,(g)\right]\right) \tag{8}$$

where $f_{\text{offset}}$ represents a lightweight convolutional network and $\Delta \in \mathbb{R}^{2\times H\times W}$ represents the predicted dense offset field. The DCN is used to spatially resample the IR shared features to achieve alignment with the RGB features

$$\mathcal{F}^{s}_{\text{ir}\to\text{rgb}}(p) = \sum_{k=1}^{K} w_k \cdot \mathcal{F}^{s}_{\text{ir}}(p + p_k + \Delta_k(p)) \tag{9}$$

where $p$ is the output position, $K$ is the kernel size of DCN, $w_k$ denotes the convolution weight of the $k$th sampling point, $p_k$ is the fixed sampling offset, and $\Delta_k(p)$ is the learned adaptive offset. This allows the network to flexibly warp the IR feature map based on cross-modal cross-correlation cues, aligning it precisely with the RGB feature map spatially. The aligned features are then further refined using a lightweight cross-attention operation to enhance semantic consistency, following the standard attention mechanism [43]. Specifically, the aligned shared feature of one modality is used to guide the semantic response of the other modality before recombination with modality-private features. Finally, the refined aligned shared features are recombined with the private features of each modality and output through residual connections

$$\mathcal{F}^{\text{out}}_{m} = \mathcal{F}^{\text{in}}_{m} + \mathcal{F}^{s}_{m,\text{cross}} + \mathcal{F}^{p}_{m} \tag{10}$$

where $\mathcal{F}^{\text{in}}_{m}$ and $\mathcal{F}^{\text{out}}_{m}$ denote the input and output features of modality $m$, respectively; $\mathcal{F}^{s}_{m,\text{cross}}$ denotes the cross-attention-refined shared feature for modality $m$; and $\mathcal{F}^{p}_{m}$ denotes the modality-private feature. This design completes the entire process from feature decoupling and spatial alignment to feature reorganization, outputting enhanced features that maintain both modality specificity and geometric consistency, laying a solid foundation for subsequent multimodal fusion operations.

### C. DCE Module

As shown in Fig. 3(e), the three aerial image multimodal object detection datasets are basically distributed in a 1:1 ratio of daytime images to nighttime images. Therefore, after completing cross-modal feature alignment, it becomes particularly important to effectively utilize the unique feature information of RGB and IR modes based on illumination intensity to ensure that the detection method is not affected by illumination and extreme weather. The information provided by the two modalities is inherently complementary: for example, as shown in Fig. 3(d), the RGB modality provides rich texture, spatial, and color details, which helps to accurately identify object categories and boundaries under good lighting conditions, while the IR modality, as shown in Fig. 3(c), can stably provide object contour and structural information based on thermal radiation, effectively overcoming the challenge of poor object recognition due to insufficient lighting or smoke occlusion. Existing methods have recognized the need to dynamically adjust fusion weights according to lighting conditions (giving higher weights to the RGB modality during the day and relying more on the IR modality at night). However, these methods only remain at the level of weighted selection of original features, failing to fully explore and enhance the unique discriminative feature components of each modality, and thus failing to fully realize the complementary potential of multimodal systems.

*1) Daytime RGB Expert Design:* Under ample daylight, RGB images offer rich texture, color, and spatial structure. To fully explore these modality-private features, we utilize spatial self-attention (SA) enhancement and design an RGB expert by specifically designing feature projection methods for different branches of query (Q), key (K), and value (V), as shown in Fig. 5(a). The Q branch is responsible for locating the texture detail regions of interest. Since the R, G, and B channels in RGB images exhibit different texture response patterns, deep convolution allows the network to learn specific texture detectors for each channel. When these Q vectors interact with subsequent K vectors, they can accurately match texture-similar regions; therefore, this is used for Q branch projection

$$Q_{\text{rgb}} = \text{DWConv}_{3\times 3}\left(F_{\text{rgb}}\right). \tag{11}$$

The K branch is responsible for providing multiscale structural references. Since objects in aerial images often have extreme aspect ratios, objects at different scales require convolutional kernels with different receptive fields to capture their contours. When calculating similarity with the Q vector, the multiscale convolutional K vector can evaluate the degree of texture matching at different structural scales, thus adaptively selecting the most relevant structural context. Therefore, it is used to extract structural information

$$K_{\text{rgb}} = \text{Concat}\left(\text{Conv}_{k\times k}\left(F_{\text{rgb}}\right)\right), k \in \{1,3,5\}. \tag{12}$$

The V branch provides color gradient information as enhancement content, while directional convolution specifically captures color gradient patterns along specific directions in natural images. Once the attention weights are determined, these directional color features are weighted and fused as enhancement content, which can strengthen the color continuity of object edges. This is crucial for distinguishing objects with similar colors but different materials, and therefore, it is used for color enhancement

$$V_{\text{rgb}} = \text{Concat}\left(\text{Conv}_{\theta}\left(F_{\text{rgb}}\right)\right), \theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\} \tag{13}$$

where Concat$(\cdot)$ denotes channel-wise concatenation over different convolutional branches. The final output from the RGB expert undergoes spatial SA and residual connection

$$F^{\text{expert}}_{\text{rgb}} = F_{\text{rgb}} + \left(\text{Softmax}\left(\frac{Q_{\text{rgb}}K^{\top}_{\text{rgb}}}{\sqrt{d}}\right)V_{\text{rgb}}\right). \tag{14}$$

The SA here establishes an association matrix between all spatial location pairs, enabling the texture pattern Q at each output location to match the global structural context K. When the local texture at one location is highly correlated with the macroscopic structure at another location, the color features of the latter are enhanced in the former, thus integrating global semantic consistency while maintaining local details.

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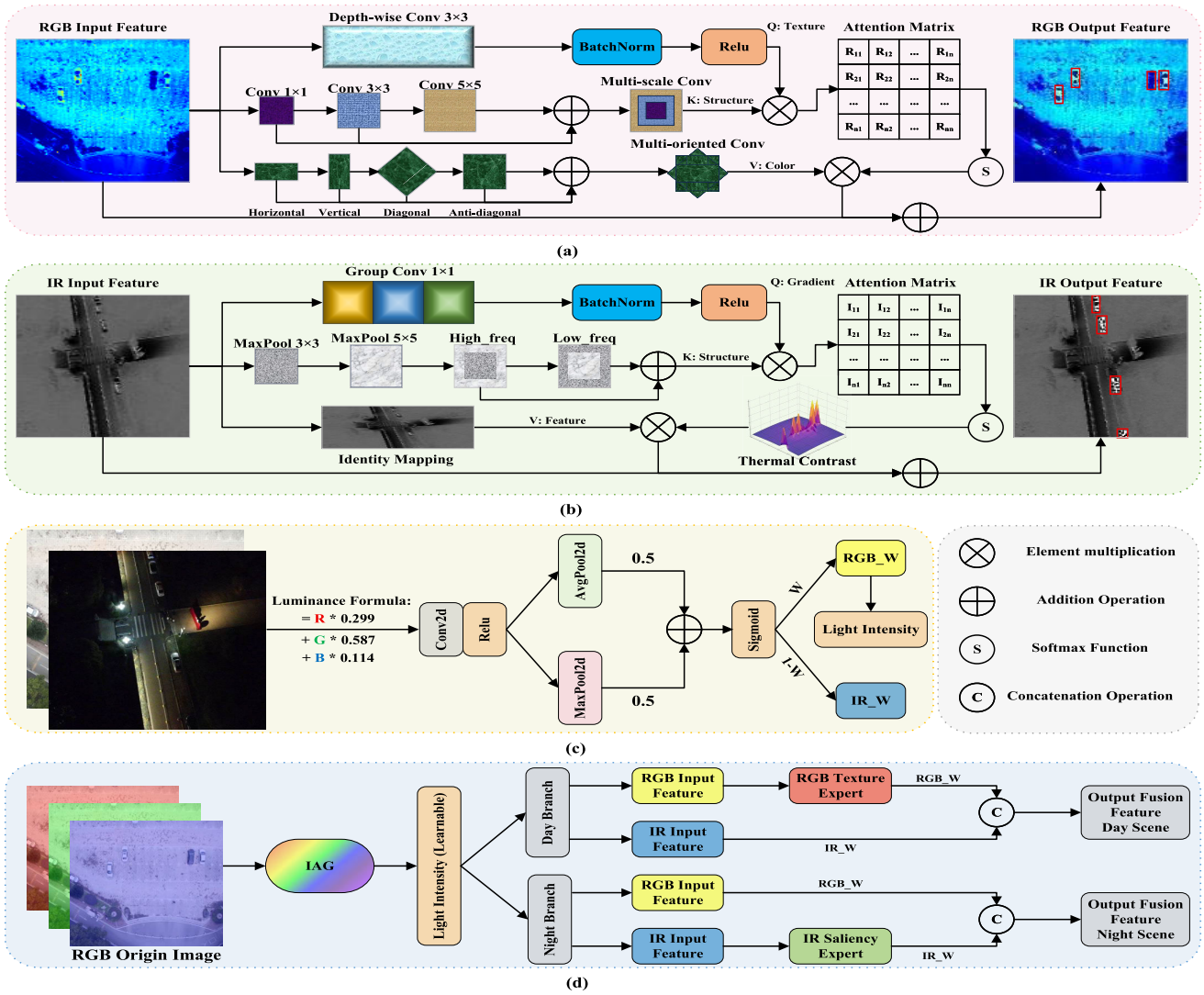



Fig. 5. Detailed process of the DCE module begins with illumination sensing via IAG, followed by targeted enhancement by RGB experts and IR experts, and finally weighted fusion. (a) Daytime RGB expert. (b) Nighttime IR expert. (c) Illumination-aware gating. (d) Dual-expert complementary enhancement module.

*2) Nighttime IR Expert Design:* Under nighttime conditions, the thermal radiation information of IR images becomes the main basis for discrimination. As shown in Fig. 5(b), the IR expert is specifically designed for thermal imaging characteristics and employs different Q, K, and V projection schemes to decompose multicomponent thermal features via spatial SA operations. Since the thermal radiation values of IR images can be roughly divided into three temperature ranges (low-temperature background, medium-temperature objects, and high-temperature heat sources), grouped convolution can divide the features into three groups according to their temperature response intensity. Each group independently learns a thermal gradient detector for that temperature range. This design allows the Q vector to distinguish thermal gradient features at different temperature levels, avoiding erroneous associations between different temperature regions. Therefore, the Q branch uses a three-grouped convolution

$$Q_{\mathrm{ir}} = \mathrm{GroupConv}_{g=3}\left(F_{\mathrm{ir}}\right). \tag{15}$$

The K branch extracts structural information of thermal objects through frequency-domain decomposition. We have designed a unified frequency decomposition formula

$$K_{\mathrm{ir}} = \mathrm{MaxPool}_{3\times3}\left(F_{\mathrm{ir}}\right) + \left(F_{\mathrm{ir}} - \mathrm{MaxPool}_{5\times5}\left(F_{\mathrm{ir}}\right)\right). \tag{16}$$

Here, low-frequency components of thermal features are extracted using $3 \times 3$ max pooling, which correspond to the main outline and temperature distribution of the thermal object. By subtracting the result of $5 \times 5$ max pooling, high-frequency components of thermal features are obtained, which encode the drastic temperature changes at the thermal boundary. Because nighttime IR detection is extremely sensitive to the authenticity of weak thermal signals, any feature transformation may distort the original thermal radiation distribution. Therefore, the V branch directly uses an identity mapping to completely preserve the original thermal radiation information

$$V_{\mathrm{ir}} = \mathrm{Identity}\left(F_{\mathrm{ir}}\right). \tag{17}$$

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The output of the IR expert follows the same operating procedure:

$$F_{\text{ir}}^{\text{expert}} = F_{\text{ir}} + \left(\text{Softmax}\left(\frac{Q_{\text{ir}}K_{\text{ir}}^{\top}}{\sqrt{d}}\right)V_{\text{ir}}\right). \tag{18}$$

The core of this expert's work lies in establishing a correlation between thermal gradient regions and thermal structure regions. When a location exhibits a significant thermal gradient pattern, the network searches for other locations with similar thermal structures and aggregates the thermal radiation information from those locations. Such a design is motivated by the common characteristics of thermal IR imagery, where the edges and inner regions of thermal targets usually exhibit consistent thermal distributions. During attention propagation, weak edge thermal signals are amplified from stable internal thermal regions while avoiding information distortion introduced by feature transformations.

*3) Illumination-Aware Gating:* To accurately assess the illumination intensity of the current image, the DCE module uses the RGB modal input image and predicts illumination information, designing the illumination-aware gating shown in Fig. 5(c). This is because the brightness of the original, unprocessed image directly reflects the actual illumination conditions of the scene, avoiding information distortion introduced by feature extraction. First, the brightness component is calculated as a proxy for illumination intensity

$$I = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \tag{19}$$

where $R, G, B \in \mathbb{R}^{H\times W}$ represent the three channels of the input RGB image, and the coefficients 0.299, 0.587, and 0.114 are determined according to the human visual system's sensitivity to the brightness of different RGB channels [44]. These coefficients are widely used in image processing and color space conversion [45], [46]. Based on the obtained brightness information, we construct spatially adaptive modality fusion weights. Considering the nonuniform distribution of illumination in real-world environments, such as fog, rain, and snow, we further employ a dynamic weight calculation strategy based on statistical normalization

$$W_{\text{rgb}} = \sigma\left(\alpha \cdot \frac{I - I_{\min}}{I_{\max} - I_{\min}} + \beta\right), \quad W_{\text{ir}} = 1 - W_{\text{rgb}} \tag{20}$$

where $I_{\max}$ and $I_{\min}$ are the minimum and maximum values of the current image brightness, respectively, and $\alpha$ and $\beta$ are learnable parameters (LPs). This formula maps brightness to the [0, 1] range through image normalization. Simultaneously, to avoid using a uniform threshold due to the diverse data distribution, it employs a learnable nonlinear transformation to calculate pixel-level fusion weights. When the brightness is close to $I_{\max}$, the RGB expert weights approach 1 and dominate; when the brightness is close to $I_{\min}$, the IR expert weights are primarily responsible, achieving fine-grained modal advantage complementarity.

Finally, as shown in Fig. 5(d), the illumination perception weights are fused with the weighted average of the two modality experts

$$F_{\text{out}} = W_{\text{rgb}} \odot F_{\text{rgb}}^{\text{expert}} + W_{\text{ir}} \odot F_{\text{ir}}^{\text{expert}} \tag{21}$$

where $\odot$ represents the element-wise multiplication. This formula dynamically adjusts the contribution ratio of RGB and IR modes according to the actual lighting conditions of each pixel, giving full play to the unique advantages of the dominant mode. By combining physics-inspired priors, data-driven modeling, and adaptive learning, DCE advances the fusion process from static feature aggregation to illumination-aware dynamic fusion, thereby improving the network's detection robustness under day-to-night and complex visual conditions.

## IV. Experiments and Analysis

### A. Datasets

*DroneVehicle:* It is a large-scale RGB-IR vehicle detection dataset [28] designed for UAV. It contains 56 878 paired visible light and IR images, covering a range of scenes from day to night. The dataset includes 28 956 daytime images, 16 986 nighttime images, and 10 936 late-night images. It provides directed bounding box annotations for 953 087 vehicle instances, categorized into five classes: Car, Truck, Bus, Van, and Freight-car. For experimentation, the dataset is divided into a training set of 17 990 image pairs, a validation set of 1469 image pairs, and a test set of 8980 image pairs.

*RGBTDronePerson:* It is the first RGB-IR dataset [47] designed for detecting tiny pedestrians on drone platforms. It contains 12 250 paired images with 70 880 labeled object instances, covering three categories: person, rider, and crowd. This dataset specifically targets the detection of small objects, with 98% of the objects being smaller than 20 pixels. It is divided into 4900 image pairs for training and 1225 image pairs for validation and testing.

*VTUAV-det:* It is an object detection dataset [47] built upon the large-scale RGB-IR UAV tracking benchmark VTUAV. It consists of one frame extracted from every 50 frames of the original video sequence, labeled only with the person category. The dataset is divided into 11 392 pairs of images as training images and 5378 pairs as validation and test images. Unlike RGBTDronePerson, which mainly contains small objects, the object scale distribution in VTUAV-det is more uniform. According to the COCO evaluation criteria, the training set contains 7651 small objects, 50 794 medium objects, and 17 869 large objects; the test set contains 9864 small objects, 32 731 medium objects, and 5960 large objects.

### B. Experimental Details and Evaluation Metrics

We design a dual-modal input and processing interface based on the single-modal 2-D detector YOLO11s [61] and adopt a dual-stream YOLO11s with mid-level concatenation as the baseline for comparison. Specifically, RGB and IR features are extracted independently through a dual-stream backbone network. Multimodal fusion is performed at three intermediate stages, namely F3–F5, where the corresponding features from both modalities are fused via channel-wise concatenation. The baseline does not introduce the COD or DCE modules. Our experimental environment consists of Python 3.10.8, Torch 2.2.2, and Ubuntu 22.04.5 LTS, using two NVIDIA RTX3090 graphics cards, each with 24 GB of VRAM. $D^2E^2$-Det is trained for 200 epochs on each of the three datasets with a

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TABLE I

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DRONEVEHICLE DATASET. THE TOP THREE RGB/IR SINGLE-MODAL AND THE MULTIMODAL METHODS (GM-DETR AND MFPT) USE RESNET-50 AS THE BACKBONE; THE OTHERS USE CSPDAKNET

| Methods | Year | Modality | Car | Truck | Freight-car | Bus | Van | $mAP_{50}$ | mAP |
|---|---|---|---|---|---|---|---|---|---|
| ROI Transformer [48] | CVPR'2019 | RGB | 81.5 | 55.0 | 42.2 | 85.4 | 44.8 | 61.5 | - |
| Oriented R-CNN [49] | ICCV'2021 | | 80.2 | 55.3 | 42.1 | 86.8 | 46.9 | 62.3 | - |
| Oriented RepPoints [50] | CVPR'2022 | | 84.4 | 55.0 | 39.5 | 85.8 | 46.6 | 62.3 | - |
| YOLOv8s-OBB [51] | GITHUB'2023 | | 91.9 | 53.1 | 56.5 | 68.8 | 92.7 | 72.6 | 45.7 |
| YOLOv12s-OBB [52] | Arxiv'2025 | | 78.7 | 63.5 | 65.3 | 94.7 | 94.4 | 79.3 | 57.8 |
| ROI Transformer [48] | CVPR'2019 | IR | 89.6 | 50.9 | 53.4 | 88.8 | 44.4 | 65.4 | - |
| Oriented R-CNN [49] | ICCV'2021 | | 89.6 | 53.9 | 53.8 | 89.1 | 40.9 | 65.5 | - |
| Oriented RepPoints [50] | CVPR'2022 | | 89.9 | 55.6 | 57.6 | 89.1 | 48.1 | 68.0 | - |
| YOLOv8s-OBB [51] | GITHUB'2023 | | 98.3 | 69.9 | 66.0 | 76.7 | 96.0 | 81.1 | 58.1 |
| YOLOv12s-OBB [52] | Arxiv'2025 | | 96.5 | 74.8 | 70.6 | 90.8 | 66.8 | 81.1 | 59.0 |
| YOLOrs [53] | JSTARS'2021 | RGB+IR | 97.7 | 77.2 | 63.6 | 65.6 | 96.0 | 80.0 | 58.1 |
| SuperYOLO [54] | TGRS'2023 | | 97.7 | 79.2 | 67.6 | 66.3 | 96.6 | 81.4 | 58.6 |
| GHOST [55] | TGRS'2023 | | 97.3 | 78.8 | 66.5 | 68.9 | 96.3 | 81.5 | 59.3 |
| MFPT [56] | TITS'2023 | | 97.3 | 72.2 | 66.7 | 77.2 | 96.6 | 82.0 | 58.7 |
| ICAFusion [57] | PR'2024 | | 96.1 | 46.4 | 34.0 | 57.1 | 92.2 | 65.1 | 44.0 |
| GM-DETR [58] | CVPR'2024 | | 92.4 | 75.3 | 64.9 | 64.9 | 80.8 | 80.5 | 55.9 |
| DaFF [59] | CVPR'2024 | | 92.2 | 58.9 | 58.2 | 71.9 | 94.4 | 75.1 | 45.5 |
| CMADet [60] | TIV'2024 | | 98.2 | 70.4 | 66.4 | 78.3 | 96.8 | 82.0 | 59.5 |
| Baseline [61] | GITHUB'2024 | | 95.4 | 76.5 | 69.6 | 70.4 | 95.2 | 81.4 | 57.9 |
| COMO [62] | IF'2025 | | 98.6 | 78.9 | 71.5 | **84.1** | **97.4** | 86.1 | 60.5 |
| $D^2E^2$-Det (Ours) | - | | **98.7** | **84.7** | **72.1** | 80.8 | 97.2 | **86.6** | **61.1** |

learning rate of $1 \times 10^{-4}$, a batch size of 4, and a weight decay of $5 \times 10^{-4}$.

For all three datasets, we use COCO-style ($mAP_{50}$) and:0.95 (mAP) as the primary metrics for comparison and analysis. In addition, for the RGBTDronePerson and VTUAV-det datasets, we follow the evaluation protocol proposed by Zhang et al. [47] for fair comparison. On RGBTDronePerson, we report results for tiny1, tiny2, tiny3, and small at an IoU threshold of 0.5, corresponding to object sizes of 0–8, 8–12, 12–20, and 20–32 pixels, respectively. On VTUAV-det, we also compute COCO-standard metrics and further analyze performance across small, medium, and large object scales.

### C. Comparisons With State-of-the-Art Models

*1) Results of DroneVehicle Dataset:* As shown in Table I, $D^2E^2$-Det outperforms other methods in both $mAP_{50}$ and mAP metrics on the DroneVehicle dataset, achieving 86.6% and 66.1%, respectively. Furthermore, all five single-modal methods lag behind the multimodal methods, indicating that single-modal approaches are insufficient for day-to-night detection, thus highlighting the need for effective multimodal aerial object detection methods. In the Truck and Freight-car categories, where distinguishing similar shapes is challenging, $D^2E^2$-Det improves by 5.8% and 9.0%, respectively, over the second-best method, COMO, from 2025. This improvement stems from the DCE module's expert enhancement mechanism, which successfully uncovers the strengths of each modality in both daytime and nighttime scenarios, thereby guiding the fusion process. In contrast, other methods, such as the simple fusion strategies in GM-DETR and standard attention mechanisms in DaFF, struggle to handle cross-modal perception and fusion from an aerial perspective.

The qualitative results on the DroneVehicle dataset are visualized in Fig. 6, comparing the $D^2E^2$-Det, $C^2$Former [13], and ICAFusion [57] methods. Overall, $D^2E^2$-Det achieves consistent and good detection results in both daytime and nighttime scenes, as well as across both RGB and IR modalities. In contrast, the other two methods exhibit false detections. This is because the RGB expert texture enhancement mechanism in the DCE module helps the model distinguish between different vehicle categories during the day, while the IR expert thermal radiation enhancement allows the model to detect vehicles obscured in the darkness.

*2) Results of RGBTDronePerson Dataset:* Table II shows the results of $D^2E^2$-Det and other state-of-the-art methods on the RGBTDronePerson dataset, which is specifically designed for small targets on UAVs. Our $D^2E^2$-Det achieves the best performance in five out of seven accuracy metrics, covering different dimensions, and comprehensively outperforms the 2025 method COXNet. Notably, the detection results of the two single-RGB modal methods are very low, indicating that the ground truth labels are based on IR modality annotations, causing a misalignment between the two modalities [as shown in Fig. 3(a)]. Therefore, effectively utilizing RGB modality information and overcoming cross-modal object offset remains a key challenge in multimodal aerial object detection. Our COD module successfully mitigates the negative impact of this annotation misalignment through orthogonal decoupling and spatial alignment mechanisms, achieving scores of 30.90%, 37.47%, and 59.54% on the highly challenging $mAP_{50}^{tiny1}$, $mAP_{50}^{tiny2}$, and $mAP_{50}^{tiny3}$ metrics, respectively. In addition, while achieving high accuracy, $D^2E^2$-Det also maintains an inference speed of 13.1 FPS and a computational cost of 70.56 GFLOPs, ranking in the middle among other methods, thus ensuring the model's feasibility for deployment on drone platforms.

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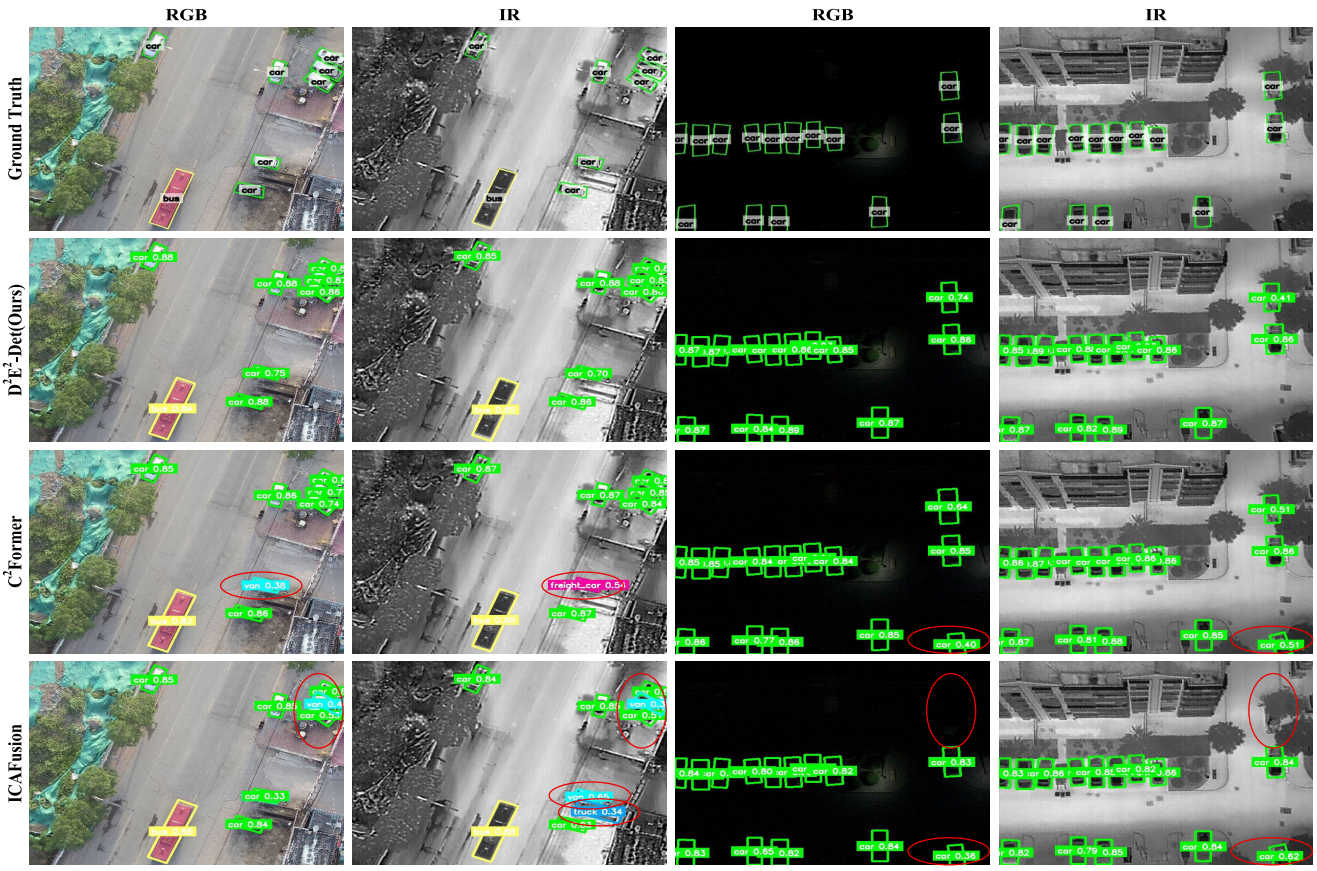



Fig. 6. Visualizations comparing our D$^2$E$^2$-Det method with other state-of-the-art methods on the DroneVehicle dataset. Results for different modalities are shown for both daytime and nighttime scenarios, with errors highlighted in red circles.

TABLE II

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE RGBTDRONEPERSON. EXCEPT FOR BASELINE AND D$^2$E$^2$-DET, ALL OTHER COMPARISON METHODS USE RESNET-50 AS THE BACKBONE. † REPRESENTS THE RESULT REPORTED BY THE RGBT BASELINE DETECTOR [47]; ‡ INDICATES OUR MMDETECTION-BASED REIMPLEMENTATIONS WITH A UNIFIED TWO-STREAM MULTIMODAL INPUT

| Methods | Paradigm | Year | Modality | $mAP_{50}^{all}$ | $mAP_{50}^{tiny}$ | $mAP_{50}^{tiny1}$ | $mAP_{50}^{tiny2}$ | $mAP_{50}^{tiny3}$ | $mAP_{50}^{small}$ | $mAP_{25}^{all}$ | FPS | GFLOPs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sparse R-CNN [63] | Two-stage | 2021 | RGB | 0.10 | 0.20 | 0.10 | 0.10 | 0.20 | 0.70 | 0.60 | 18.3 | 122.00 |
| DINO [64] | DETR-series | 2023 | | 0.20 | 0.40 | 0.10 | 0.30 | 0.03 | 0.70 | 1.10 | 19.8 | 211.00 |
| Sparse R-CNN [63] | Two-stage | 2021 | IR | 24.10 | 19.60 | 5.90 | 11.80 | 15.70 | 10.50 | 31.40 | 18.3 | 122.00 |
| DINO [64] | DETR-series | 2023 | | 40.80 | 32.90 | 9.80 | 19.50 | 26.70 | 19.70 | 51.90 | 19.8 | 211.00 |
| Cascade R-CNN† [65] | Two-stage | 2018 | RGB+IR | 31.55 | 31.99 | 0.00 | 29.43 | 37.77 | 33.61 | 42.47 | 11.8 | 76.23 |
| FCOS† [66] | One-stage | 2019 | | 29.89 | 30.71 | 9.40 | 22.73 | 34.87 | 26.00 | 45.21 | 19.2 | 47.21 |
| ATSS† [67] | One-stage | 2020 | | 36.24 | 37.47 | 16.92 | 24.16 | 43.59 | 24.62 | 53.14 | 19.0 | 48.73 |
| GFL† [68] | One-stage | 2020 | | 38.77 | 40.00 | 17.86 | 28.15 | 46.66 | 26.75 | 55.59 | 18.7 | 49.22 |
| CFT [69] | One-stage | 2021 | | 22.69 | 22.83 | 16.72 | 18.14 | 27.52 | 8.14 | 37.32 | 10.8 | 112.02 |
| ProbEn [70] | Two-stage | 2022 | | 3.64 | 3.83 | 0.77 | 5.41 | 5.19 | 0.33 | 10.73 | - | - |
| HRFuser [71] | Two-stage | 2022 | | 22.23 | 22.50 | 0.00 | 26.73 | 26.26 | 23.85 | 33.24 | 4.5 | 54.17 |
| DINO‡ [64] | DETR-series | 2022 | | 41.71 | 33.61 | 12.39 | 24.78 | 37.95 | 21.05 | 48.26 | 6.8 | 126.00 |
| TINet [17] | Two-stage | 2023 | | 28.30 | 28.60 | 0.00 | 24.12 | 34.99 | **34.97** | 40.34 | 13.0 | 92.80 |
| QFDet [47] | One-stage | 2023 | | 42.08 | 44.04 | 20.27 | 30.09 | 50.36 | 26.78 | 57.34 | 14.2 | 81.43 |
| Baseline [61] | One-stage | 2024 | | 41.18 | 42.87 | 19.41 | 30.36 | 50.27 | 26.73 | 54.48 | 20.4 | 58.23 |
| COXNet [72] | One-stage | 2025 | | 45.57 | 47.18 | 27.37 | 35.55 | 52.56 | 29.74 | **59.01** | 17.6 | 51.27 |
| D$^2$E$^2$-Det (Ours) | One-stage | - | | **47.72** | **49.89** | **30.90** | **37.47** | **59.54** | 28.04 | 52.11 | 13.1 | 70.56 |

The qualitative results of training three multimodal methods on the RGBTDronePerson dataset and predicting RGB modal images are shown in Fig. 7. To more clearly demonstrate the advantages of D$^2$E$^2$-Det compared to other state-of-the-art methods, we use the ground truth labels as a reference, define a correct detection as having an IoU threshold greater than 0.4, and represent it with green boxes. False positives are marked with red boxes, and false negatives are indicated with blue boxes. Overall, D$^2$E$^2$-Det achieves the most balanced performance across different lighting scenarios, with only one false negative and one false positive, and it remains robust even under the challenging nighttime overexposure condition. QFDet [47] performs second best, with approximately the same number of green boxes and blue + red boxes. This is due

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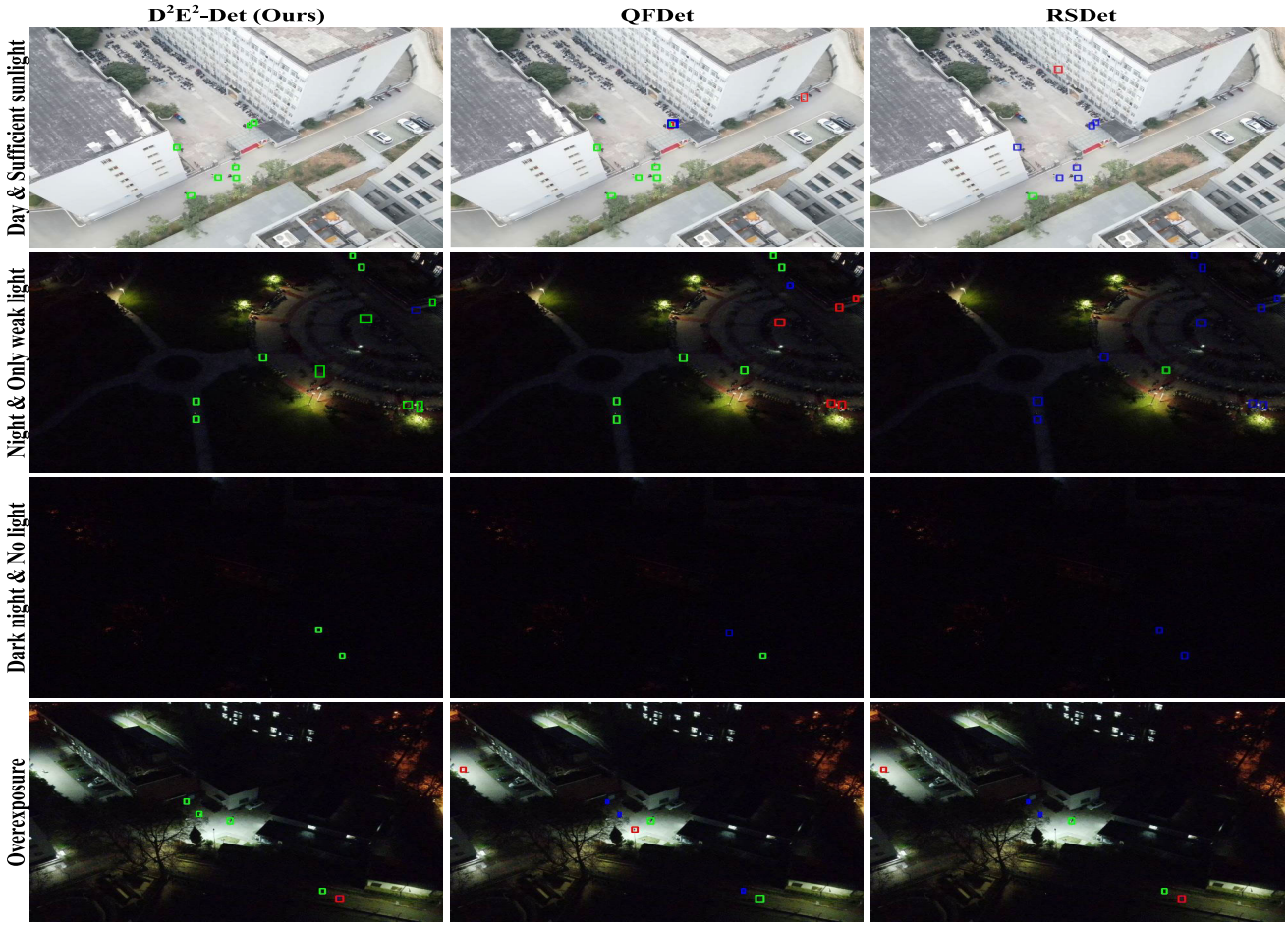



Fig. 7. This visualization compares the results of our D$^2$E$^2$-Det method on the RGBTDronePerson dataset for the RGB modal with other state-of-the-art methods. Daytime, night, and dark night scenes are compared; green boxes represent correct detections, red boxes represent false positives, and blue boxes represent false negatives.

TABLE III

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE VTUAV-DET DATASET. EXCEPT FOR BASELINE AND D$^2$E$^2$-DET, ALL OTHER COMPARISON METHODS USE RESNET-50 AS THE BACKBONE

| Methods | Year | Modality | mAP | $mAP_{50}$ | $mAP_{75}$ | $mAP_s$ | $mAP_m$ | $mAP_l$ | Param |
|---|---|---|---|---|---|---|---|---|---|
| SSPNet [74] | GRSL'2021 | RGB | 12.00 | 39.90 | 3.70 | 3.60 | 11.60 | 26.20 | 46.79 |
| DotD [75] | CVPR'2021 | | 12.20 | 40.50 | 3.70 | 3.70 | 11.50 | 27.30 | 41.13 |
| NWD-PKA [76] | ISPRS'2022 | | 12.40 | 40.70 | 3.80 | 4.10 | 11.50 | 27.80 | 41.13 |
| RT-DETR [77] | CVPR'2024 | | 13.41 | 43.44 | 4.60 | 4.35 | 11.87 | 27.62 | 19.87 |
| SSPNet [74] | GRSL'2021 | IR | 31.60 | 71.30 | 22.70 | 16.50 | 31.00 | 55.10 | 46.79 |
| DotD [75] | CVPR'2021 | | 32.00 | 71.90 | 23.30 | 17.40 | 31.00 | 55.30 | 41.13 |
| NWD-RKA [76] | ISPRS'2022 | | 31.60 | 72.50 | 22.20 | 17.10 | 30.70 | 55.00 | 41.13 |
| RT-DETR [77] | CVPR'2024 | | 33.07 | 74.20 | 24.80 | 19.07 | 33.10 | 57.14 | 19.87 |
| UA-CMDet [28] | TCSVT'2022 | RGB+IR | 32.25 | 70.05 | 24.86 | 14.93 | 32.47 | 57.76 | 138.70 |
| QueryDet [78] | CVPR'2022 | | 29.50 | 68.90 | 20.20 | 7.80 | 29.80 | 53.50 | 62.58 |
| RSDet [73] | Arxiv'2024 | | 29.80 | 72.10 | 18.50 | 16.50 | 30.70 | 44.60 | 166.66 |
| TFDet [79] | TNNLS'2024 | | 26.50 | 64.20 | 16.80 | 9.20 | 28.50 | 44.30 | 169.33 |
| Baseline [61] | GITHUB'2024 | | 31.43 | 73.09 | 22.95 | 16.54 | 31.08 | 53.89 | 32.26 |
| AMSDet [80] | TIM'2025 | | 33.60 | 75.70 | 24.20 | 18.10 | 33.40 | 56.20 | 50.01 |
| COXNet [72] | TCSVT'2025 | | 33.50 | 76.10 | 25.10 | 18.60 | 32.60 | 56.80 | 71.11 |
| D$^2$E$^2$-Det (Ours) | - | | **35.09** | **78.46** | **28.23** | **21.77** | **36.47** | **58.05** | 39.44 |

to QFDet's fusion strategy, which fails to address the cross-modal feature misalignment problem, resulting in a decrease in its quality perception ability when relying on thermal radiation priors during RGB modal inference. RSDet [73] performs worst, with approximately 85% false negatives. This is because the coarse-grained filtering of its RSR module in the frequency domain may inadvertently damage small target features in the RGB modality, while the DFS module's selection mechanism

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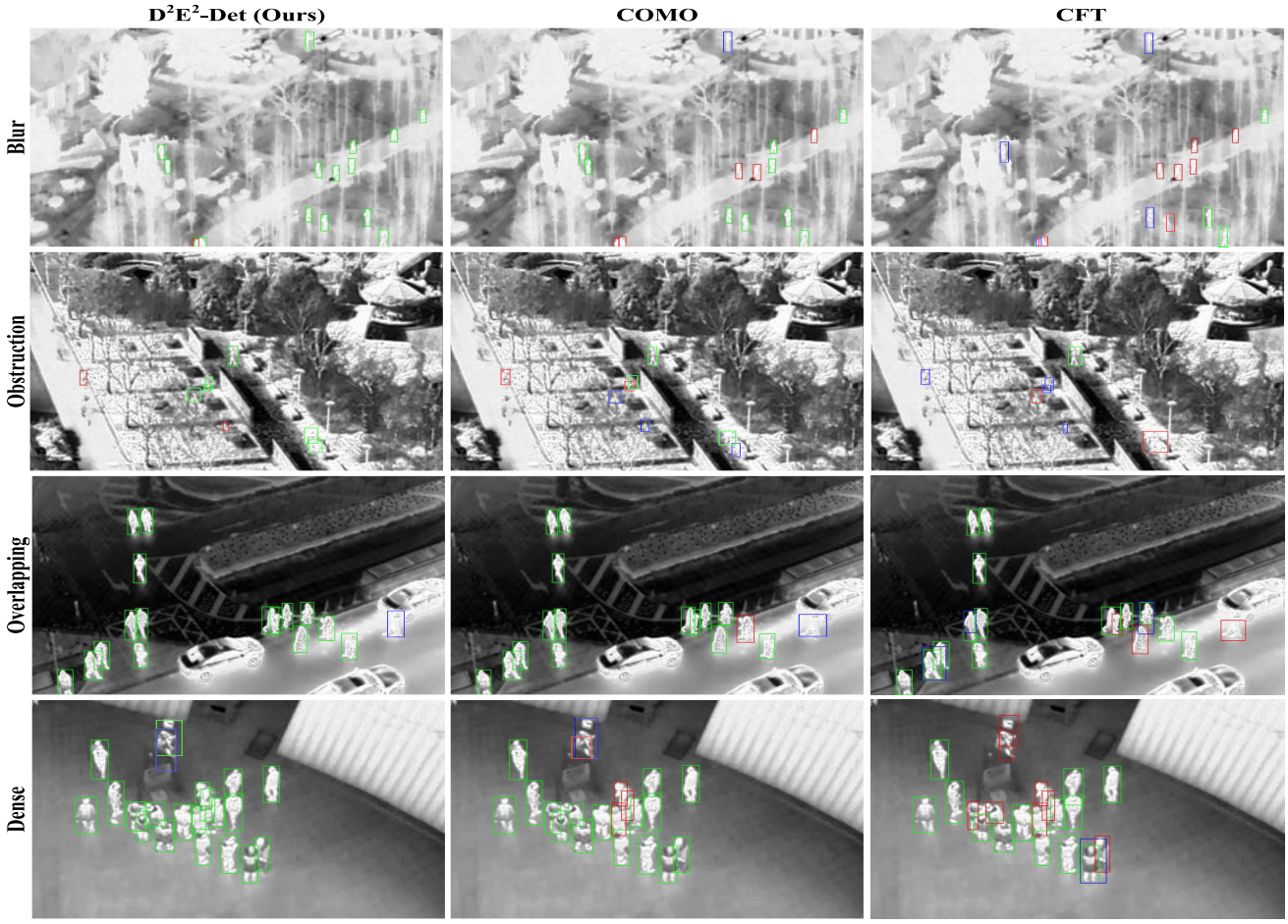



Fig. 8. This visualization compares the results of our D$^2$E$^2$-Det method on the VTUAV-det dataset for the IR modal with other state-of-the-art methods. Results are presented under four extremely challenging aerial perspective scenarios (blurred, occluded, overlapping, and dense), with green boxes representing correct detections, red boxes representing false positives, and blue boxes representing missed detections.

fails when features are severely misaligned. In contrast, D$^2$E$^2$-Det separates shared and private features through the COD module and leverages DCE to adaptively enhance modality-specific cues under varying illumination conditions. Under challenging nighttime overexposure, strong local saturation may affect illumination perception in the IAG module, which can further propagate to DCE and result in occasional semantic ambiguity and misclassification errors.

*3) Results of VTUAV-Det Dataset:* Table III shows the performance comparison of D$^2$E$^2$-Det with four other single-modal methods and seven multimodal methods on the VTUAV-det dataset for pedestrian detection from a UAV perspective. Our method achieves the best results in all six accuracy metrics, improving upon the second-best results by 1.59%, 2.36%, 3.13%, 2.70%, 3.07%, and 0.29%. Furthermore, D$^2$E$^2$-Det not only effectively locates small objects, thereby improving mAP$_s$, but also accurately identifies large objects to maintain the advantage of the mAP$_l$ score. This is mainly due to the COD module's strategy of decoupling and aligning different modal features, which effectively corrects geometric deformations during the dynamic shooting process of the UAV and provides the detection head with high-confidence and accurate feature representations. Finally, it is worth noting that while achieving the best performance, D$^2$E$^2$-Det requires only 39.44 M parameters, fewer than other state-of-the-art multimodal methods. This proves that our COD and DCE not only do not increase model complexity but also achieve a dual guarantee of accuracy and parameter efficiency through careful and reasonable design, providing a feasible solution for day-to-night aerial object detection when deployed on UAVs.

To more comprehensively analyze the performance of D$^2$E$^2$-Det from an aerial perspective, we specifically select four highly challenging scenarios (blur, obstruction, overlapping, and dense) from the VTUAV-det dataset and compare the visualization results of our method with two other state-of-the-art multimodal methods in the IR modality, as shown in Fig. 8. Similar to Fig. 7, we use three colored anchor boxes to more intuitively distinguish the detection results. Overall, in the four extreme scenarios, D$^2$E$^2$-Det achieves the vast majority of correct detections, with only 1–2 false positives or false negatives in each scenario, demonstrating the strongest robustness and the most accurate localization capability. This superiority mainly stems from our DCE's ability to use targeted experts to capture and enhance the original semantic information of objects in different weather conditions, while COD ensures that features from another modality can achieve accurate alignment during complementary fusion, achieving a

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TABLE IV

ABLATION EXPERIMENTS OF DIFFERENT MODULES IN D$^2$E$^2$-DET ON THREE DATASETS

| COD | DCE | DroneVehicle | | RGBTDronePerson | | VTUAV-det | |
|---|---|---|---|---|---|---|---|
| | | $mAP_{50}$ | mAP | $mAP_{50}$ | mAP | $mAP_{50}$ | mAP |
| | | 81.42 | 57.89 | 41.18 | 15.33 | 73.09 | 31.43 |
| ✓ | | 83.78 | 60.28 | 44.23 | 17.21 | 75.78 | 33.26 |
| | ✓ | 83.49 | 59.84 | 44.30 | 16.90 | 75.81 | 33.22 |
| ✓ | ✓ | **86.60** | **61.10** | **47.72** | **19.42** | **78.46** | **35.09** |

TABLE V

COMPONENT ABLATION OF COD MODULE AND ALIGNMENT METHOD COMPARISON, DEMONSTRATING THE RESULTS OF $mAP_{50}$

| Methods | DroneVehicle | RGBTDronePerson | VTUAV-det |
|---|---|---|---|
| No | 83.49 | 44.30 | 75.81 |
| w/ Alignment | 83.79 | 46.08 | 76.92 |
| w/ Decouple | 85.41 | 46.71 | 77.02 |
| w/o Orth. Gate | 85.10 | 44.90 | 76.54 |
| DCN [81] | 84.46 | 46.28 | 75.82 |
| CA [43] | 85.12 | 46.47 | 76.29 |
| CLC [14] | 86.08 | 47.01 | 77.28 |
| CAGT [11] | 85.51 | 46.77 | 76.44 |
| COD | **86.60** | **47.72** | **78.46** |

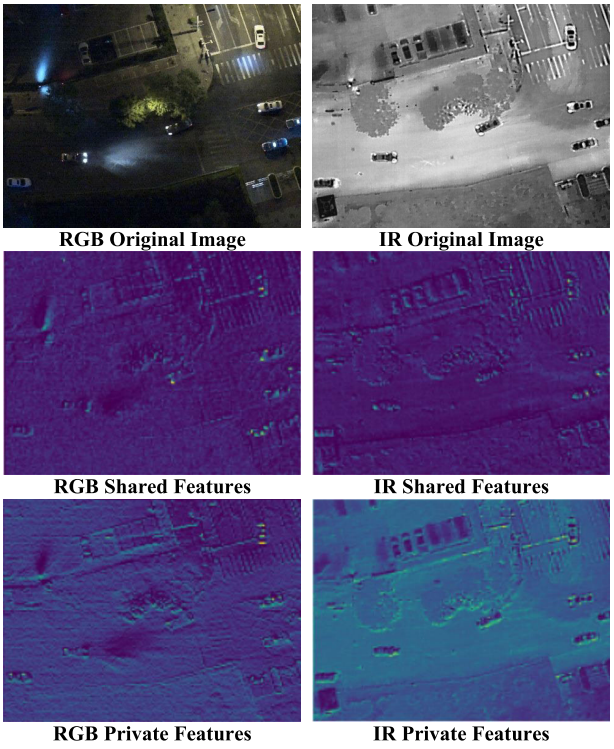



Fig. 9. Visualization of shared and private features decoupled by the proposed COD module.

$1 + 1 > 2$ effect. In contrast, although COMO [62] achieves efficient interaction through the Mamba model, its strategy of using only high-level feature fusion loses a lot of detailed information, and its offset guidance mechanism struggles to dynamically adapt to feature deformations in complex scenarios. Similarly, although CFT [69] uses the Transformer model for global fusion, its soft alignment mechanism cannot solve the problem of geometric misalignment between modalities caused by sensor differences. It is more likely to introduce spatial noise in difficult and complex scenes, which limits the detection performance from an aerial perspective.

### D. Ablation Studies

*1) Effectiveness of Different Components in D$^2$E$^2$-Det:* To verify the effectiveness of the two components integrated in D$^2$E$^2$-Det, we conduct ablation studies on three datasets, and the results are shown in Table IV. Both COD and DCE, used independently, provide performance gains on all three datasets, while the combination of the two achieves optimal performance across all metrics, validating the effectiveness and necessity of each component. Specifically, on the three datasets, the COD + DCE combination improves the performance by 5.18%, 3.21%, 6.54%, 4.09%, 5.37%, and 3.66% on six metrics compared to the baseline. This indicates that the clean, decoupled features provided by the COD module lay a solid foundation for more accurate feature selection and enhancement by experts in the DCE module. The two components complement and promote each other, jointly driving the upper limit of the model's detection performance in complex day-to-night scenarios.

*2) Effectiveness Experiment Within the COD Module:* As shown in Table V, we compare the core operation of the COD module with several other cross-modal alignment methods at the cross-modal feature alignment stage, namely layers F3–F5 of the dual-stream backbone network, and report the $mAP_{50}$ results. The results show that COD achieves the best performance on all three datasets, benefiting from its innovative decoupling first, alignment later paradigm. Specifically, the No setting yields the lowest performance, directly demonstrating the necessity of cross-modal feature alignment. Applying only alignment (w/Alignment) or only decoupling (w/Decouple) brings only limited improvements while removing the orthogonal-gate operation (w/o Orth. Gate), which removes the cosine-similarity-guided complementary masks $G_m$ and $1 - G_m$ while retaining the disentanglement, spatial alignment, and cross-attention operations, leads to performance degradation, which verifies the critical role of orthogonal decoupling in preserving the purity of shared and private features. The limited local transformation capability of DCN and the restricted global interaction of CA suggest that simple geometric modeling or soft alignment is insufficient to address complex spatial offset issues. In contrast, although CLC and CAGT achieve improvements through explicit calibration modules, they still operate on mixed features without distinguishing between shared and modality-specific components, making them more vulnerable to intramodal noise. Finally, COD performs alignment on the pure shared features obtained through orthogonal decoupling, which not only avoids interference from noisy source features but also enables the alignment process to focus more precisely on geometrically consistent information.

As shown in Fig. 9, the COD module effectively separates the shared and private features of the RGB-IR modalities

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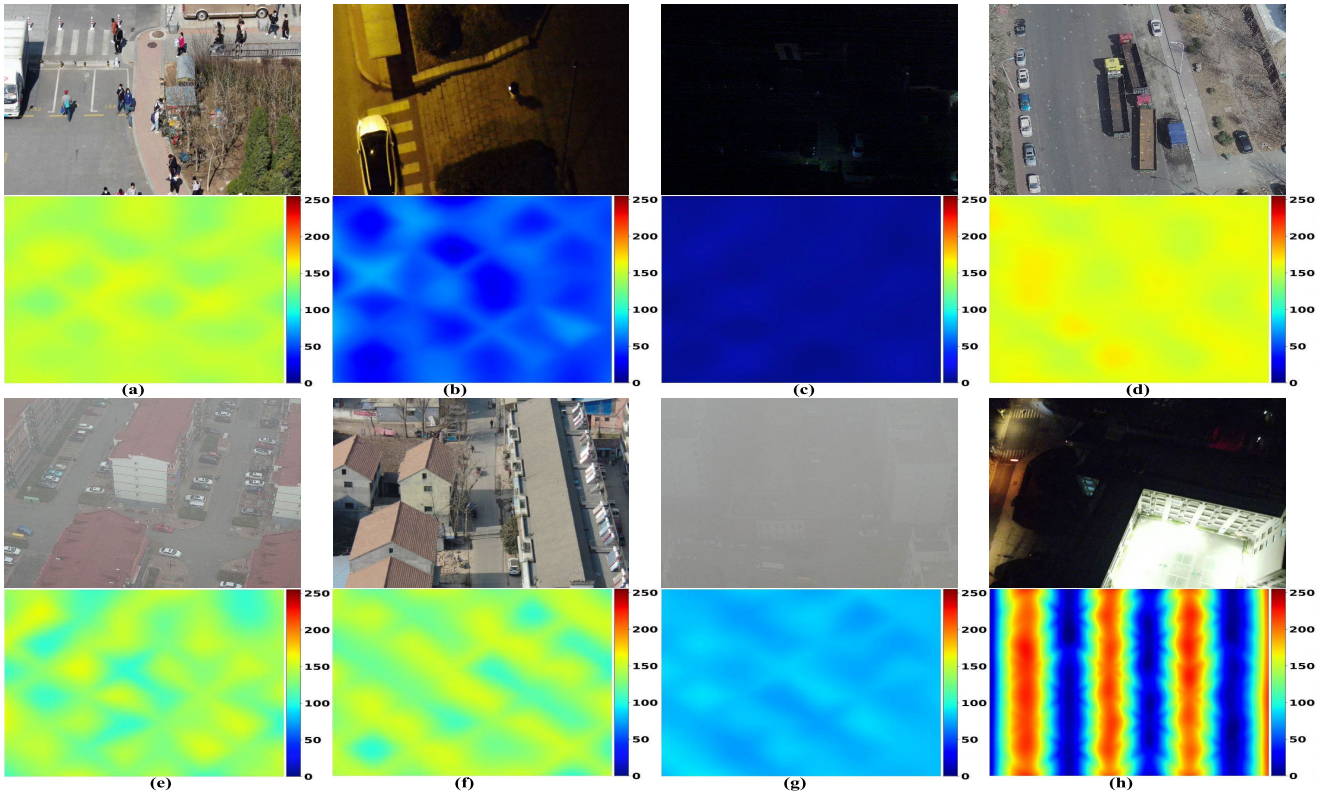



Fig. 10. Visualization of RGB responses of IAG under eight representative illumination scenarios. (a) Day. (b) Night. (c) Dark night. (d) Snow. (e) Haze. (f) Shadow. (g) Overcast. (h) Overexposure.

TABLE VI
COMPARISON OF DIFFERENT PROJECTION METHODS WITHIN DCE MODULE AND STANDARD ATTENTION MODULES, DEMONSTRATING THE RESULTS OF $MAP_{50}$

| Methods | DroneVehicle | RGBTDronePerson | VTUAV-det |
|---|---|---|---|
| w/ IA | 84.64 | 45.98 | 76.63 |
| IA + SE [82] | 84.41 | 46.02 | 76.71 |
| IA + CBAM [83] | 84.59 | 45.10 | 76.97 |
| IA + Swin [84] | 85.07 | 45.45 | 77.12 |
| IA + SA (Conv) | 84.97 | 45.78 | 77.05 |
| IA + SA (Linear) | 85.02 | 46.54 | 76.88 |
| DCE | **86.60** | **47.72** | **78.46** |

through orthogonal decoupling. The shared features in the second row accurately preserve the target structures common across modalities, such as vehicles, thereby providing pure geometric consensus information for subsequent alignment. In contrast, the private features in the third row capture modality-specific information, such as RGB texture details of vehicles and IR thermal radiation noise. This demonstrates that the proposed module effectively separates useful modality-invariant features from modality-specific interference, thereby verifying the effectiveness of the decoupling mechanism.

*3) Effectiveness Experiment Within the DCE Module:* To verify the effectiveness of the DCE module design, we conduct ablation experiments on its core enhancement components, and the results are presented in Table VI. Specifically, we compare the baseline scheme using only IA, variants that incorporate SE, CBAM, and Swin attention modules on top of IA, and enhancement schemes with different SA projection strategies, thereby comprehensively validating the performance advantages of the proposed DCE. The results show that the baseline with only IA achieves the lowest performance, which highlights the necessity of modality-specific enhancement. Although SE, CBAM, and Swin each exhibit certain advantages on different datasets, their overall performance remains inferior to that of DCE, indicating that general attention mechanisms are not well suited to the characteristic differences between RGB and IR modalities. Moreover, although IA+SA (Conv/Linear) introduces SA, its performance is still inferior to that of DCE and shows noticeable fluctuations, suggesting that it cannot stably exploit modality-specific features. Ultimately, the proposed DCE achieves the best performance on all three datasets, surpassing the IA baseline by 1.96%, 1.74%, and 1.83%. These results validate that the dual-expert structure, which is specifically designed according to the imaging characteristics of RGB and IR modalities, provides specialized enhancement pathways for different modalities, effectively mines private discriminative features, and enables robust detection under diverse illumination conditions.

To analyze whether IAG provides effective illumination weights under different illumination and visual-degradation scenarios, we select eight representative lighting conditions and statistically visualize the RGB responses generated by IAG. A heatmap ranging from 0 to 255 is used to characterize the relative strength of the RGB brightness response, where higher values indicate a greater contribution assigned to the RGB modality. As shown in Fig. 10, IAG operates stably under common illumination changes. In daytime, evening, and

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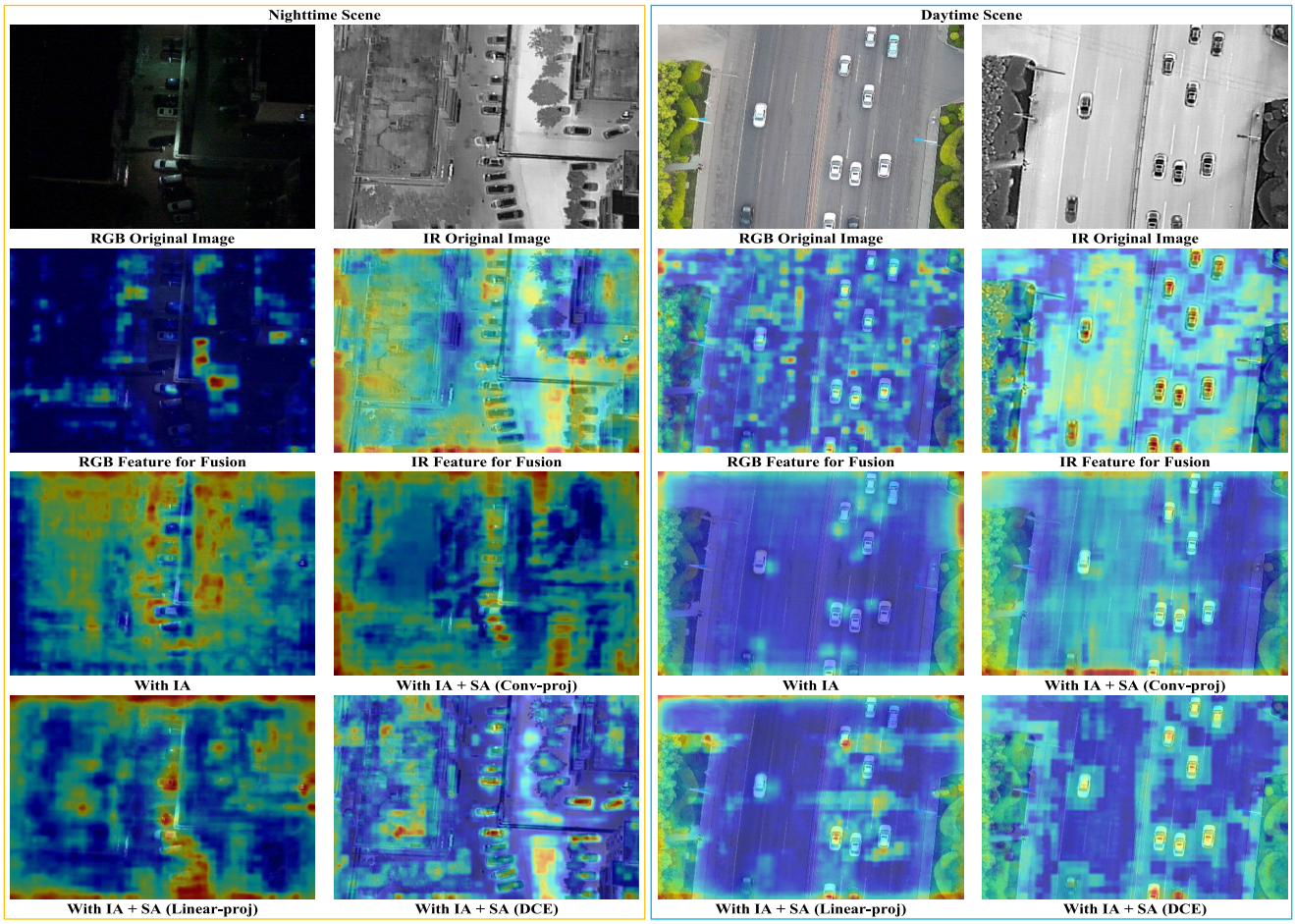



Fig. 11. Visualization results of the ablation experiment heatmap within the DCE module show both daytime and nighttime scenes. During fusion, three layers (P3–P5) from both modalities are used as input features.

late-night scenes, the RGB weights respond consistently to overall brightness variations. In scenes with snow, light haze, daytime shadows, and severe cloud cover, IAG also provides effective weighting references. This is because snow does not destroy RGB details, light haze mainly reduces brightness and contrast without altering the overall illumination structure, shadows usually affect only local regions, and severe cloud cover weakens the RGB modality and makes the IR modality more important. In contrast, in nighttime overexposure scenes with strong glare, localized highlights significantly increase the RGB brightness response, but these areas are often accompanied by detail saturation and texture loss, which reduces the reliability of brightness as an indicator of information quality and leads to unstable IAG judgments. This limitation may be further alleviated in future work through a region-based local perception strategy.

To further investigate how the dual-expert enhancement mechanism in DCE affects cross-modal fusion, we visualize and analyze the heatmaps in Table VI using different fusion schemes. As shown in Fig. 11, the first row displays the RGB and IR input images for daytime and nighttime scenes, the second row shows the input features of RGB and IR before cross-modal fusion (affected by layers F3–F5), and the third and fourth rows present the feature activation results after cross-modal fusion. When using only IA for fusion, the feature response is relatively diffuse in the target region, and the background noise is significant, indicating that the simple gating weighting strategy is insufficient to focus and enhance the discriminative features within the modality. However, after introducing SA enhancement, the feature focus improves for both Conv projection and Linear projection, and simple background noise (such as roads) is suppressed to some extent in daytime scenes. However, in nighttime scenes, the SA method also activates trees that are similar in shape to vehicles and difficult to distinguish, indicating that the general SA mechanism lacks the ability to process modality-private physical characteristics and struggles to effectively handle nighttime scene detection. In contrast, DCE demonstrates excellent feature focusing capabilities both day and night, with heatmaps showing high accuracy and clear outlines of the target area, while also exhibiting good suppression of complex background interference.

Furthermore, we conduct experiments on three datasets regarding whether to use an LP strategy or a fixed threshold for the IAG in the DCE module, as shown in Fig. 12, comparing the trends of $mAP_{50}$ and mAP metrics. The experimental results show that LP generally performs best. However, based on the information in Fig. 3(e), the effect of the fixed thresh-

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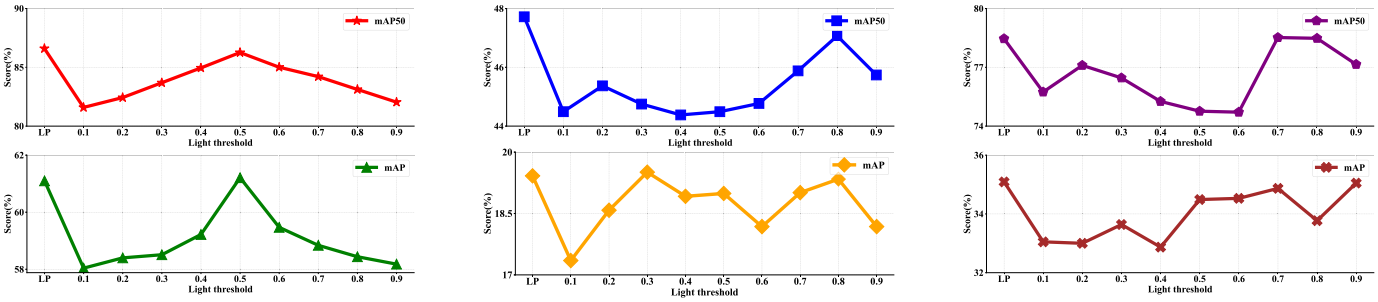



Fig. 12. Performance of IAG with Fixed and "LP." The first row presents $mAP_{50}$ results, and the second row shows mAP results. From left to right, the three columns correspond to results on the DroneVehicle, RGBTDronePerson, and VTUAV-det datasets.

old exhibits a certain degree of correlation with the scene distribution of the dataset. On the DroneVehicle dataset, the daytime and nighttime sample distribution is relatively uniform, and the fixed threshold at 0.5 achieves performance similar to or even better than LP, with performance decreasing toward both sides of 0.5. On the RGBTDronePerson dataset, samples are mainly concentrated in the two extreme conditions of daytime and late night, with a lower proportion in the evening. This results in generally low performance for the fixed threshold in the middle, and the overall line graph trend fluctuates greatly, lacking a stable pattern. On the VTUAV-det dataset, daytime samples account for the vast majority, so the fixed threshold performs relatively well in the range biased toward daytime fusion (0.7–0.9), approaching the performance of LP. The above phenomena indicate that fixed threshold strategies rely on the sample illumination distribution of the dataset and are difficult to adapt to complex and ever-changing scenarios, limiting their potential for real-time applications. In contrast, LP, through dynamic learning, responds more robustly to the actual input illumination conditions, thereby demonstrating stable and leading performance across diverse datasets.

### E. General Verification

To further validate the universality of $D^2E^2$-Det beyond the YOLO architecture, we reimplement several representative detectors in MMDetection under a unified two-stream multimodal setting, including the one-stage DDOD, the two-stage Sparse R-CNN, and the Transformer-based DINO. Following the original design of $D^2E^2$-Det, COD and DCE are consistently inserted into the two-stream backbone for multimodal feature interaction, while the subsequent neck or Transformer encoder and detection head retain the original architecture of the corresponding detector.

As shown in Table VII, after introducing $D^2E^2$-Det, DDOD improves its $mAP_{50}$ by 5.11% and 4.65% on the two datasets. Similarly, Sparse R-CNN achieves gains of 6.28% and 4.60%. These consistent improvements indicate that the proposed COD and DCE modules are compatible with both one- and two-stage CNN-based detectors. This is because both modules operate on convolutional feature maps, where local spatial structures, translation consistency, and pyramid-level representations are explicitly preserved. Therefore, the decoupling-before-alignment strategy and the dual-expert enhancement mechanism can directly improve the quality of multimodal features before detection.

TABLE VII

EXPERIMENTS ON THE UNIVERSALITY OF $D^2E^2$-DET UNDER DIFFERENT DETECTION PARADIGMS, ‡ INDICATES OUR MMDETECTION-BASED REIMPLEMENTATIONS WITH A UNIFIED TWO-STREAM MULTIMODAL INPUT

| Methods | RGBTDronePerson | VTUAV-det |
|---|---|---|
| DDOD‡ [85] | 35.67 | 68.60 |
| +$D^2E^2$-Det | **40.78**$_{+5.11}$ | **73.25**$_{+4.65}$ |
| Sparse R-CNN‡ [63] | 34.23 | 67.27 |
| +$D^2E^2$-Det | **40.51**$_{+6.28}$ | **71.87**$_{+4.60}$ |
| DINO‡ [64] | 41.71 | 74.02 |
| +$D^2E^2$-Det | **42.90**$_{+1.19}$ | 73.54 |

For the Transformer-based DINO, the $mAP_{50}$ improves by 1.19% on the RGBTDronePerson dataset, but slightly decreases on the VTUAV-det dataset. This relatively limited improvement may be attributed to the architectural mismatch between $D^2E^2$-Det and Transformer-based detectors. Specifically, DINO relies on global query interaction and set-based prediction, in which object queries aggregate long-range semantic information after feature encoding. In this case, the local alignment and modality-specific enhancement introduced by COD and DCE may be partially absorbed or weakened by the subsequent global query modeling process. In addition, the inserted convolution-oriented interaction modules may disturb the feature distribution expected by the Transformer encoder and decoder, especially on VTUAV-det, where the object scales and scene layouts are more diverse.

## V. CONCLUSION

This article proposed a multimodal detection method, termed $D^2E^2$-Det, for day-to-night aerial images. The proposed method effectively addressed the challenges of cross-modal geometric noise interference and insufficient utilization of modality-private discriminative power through a collaborative design of decoupling followed by enhancement. Specifically, the COD module explicitly separated modality-shared and modality-private features via orthogonality constraints, ensuring the purity of spatial alignment, while the DCE module enabled deep mining and complementary fusion of discriminative modal features through an illumination-aware gating mechanism and dedicated expert networks.

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Experiments conducted on three publicly available aerial multimodal datasets demonstrated that $D^2E^2$-Det outperformed existing mainstream methods in terms of detection accuracy while maintaining a favorable balance between model complexity and inference efficiency, indicating strong potential for practical deployment. Future work will investigate the extension of this framework to video-based detection and fusion tasks involving additional modalities, such as SAR and hyperspectral images.

## REFERENCES


[1] X. Wang, H. Chen, X. Chu, and P. Wang, "AODet: Aerial object detection using transformers for foreground regions," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–11, 2024.

[2] H. He, J. Ding, B. Xu, and G.-S. Xia, "On the robustness of object detection models on aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–12, 2025.

[3] H. Luo, G. Li, D. Zou, K. Li, X. Li, and Z. Yang, "UAV navigation with monocular visual inertial odometry under GNSS-denied environment," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023.

[4] X. Jiang, B. Huang, and Y. Zhao, "Spatiotemporal image fusion with spectrally preserved pre-prediction: Tackling complex land-cover changes," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024.

[5] J. Guo, Y. Wan, A. Ma, and Y. Zhong, "Disaster-aware path planning based on reinforcement learning for postearthquake emergency response," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–14, 2025.

[6] X. Zhang, Y. Feng, N. Wang, G. Lu, and S. Mei, "Transformer-based person detection in paired RGB-T aerial images with VTSaR dataset," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5082–5099, 2025.

[7] X. Zhang, Y. Feng, N. Wang, G. Lu, and S. Mei, "Aerial person detection for search and rescue: Survey and benchmarks," *J. Remote Sens.*, vol. 5, p. 474, Jan. 2025.

[8] C. Li et al., "Detection-friendly dehazing: Object detection in real-world hazy scenes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8284–8295, 2023.

[9] H. Zhou et al., "Remote sensing image dehazing: A systematic review of progress, challenges, and prospects," *ISPRS J. Photogramm. Remote Sens.*, vol. 235, pp. 318–348, May 2026.

[10] H. Tang, Z. Li, Z. Peng, and J. Tang, "BlockMix: Meta regularization and self-calibrated inference for metric-based meta-learning," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 610–618.

[11] M. Yuan, X. Shi, N. Wang, Y. Wang, and X. Wei, "Improving RGB-infrared object detection with cascade alignment-guided transformer," *Inf. Fusion*, vol. 105, May 2024, Art. no. 102246.

[12] J. Xie, J. Nie, B. Ding, M. Yu, and J. Cao, "Cross-modal local calibration and global context modeling network for RGB–infrared remote-sensing object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8933–8942, 2023.

[13] M. Yuan and X. Wei, "C2former: Calibrated and complementary transformer for RGB-infrared object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024.

[14] Y. Liu, W. Guo, C. Yao, and L. Zhang, "Dual-perspective alignment learning for multimodal remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–15, 2025.

[15] K. Chen, J. Liu, and H. Zhang, "IGT: Illumination-guided RGB-T object detection with transformers," *Knowl.-Based Syst.*, vol. 268, May 2023, Art. no. 110423.

[16] Q. Wang, Y. Chi, T. Shen, J. Song, Z. Zhang, and Y. Zhu, "Improving RGB-infrared object detection by reducing cross-modality redundancy," *Remote Sens.*, vol. 14, no. 9, p. 2020, Apr. 2022.

[17] Y. Zhang, H. Yu, Y. He, X. Wang, and W. Yang, "Illumination-guided RGBT object detection with inter- and intra-modality fusion," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.

[18] J. Wu, T. Shen, Q. Wang, Z. Tao, K. Zeng, and J. Song, "Local adaptive illumination-driven input-level fusion for infrared and visible object detection," *Remote Sens.*, vol. 15, no. 3, p. 660, Jan. 2023.

[19] X. Zhang, Y. Feng, S. Zhang, N. Wang, G. Lu, and S. Mei, "Robust aerial person detection with lightweight distillation network for edge deployment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024.

[20] L. Zhou, S. Zhao, Z. Wan, Y. Liu, Y. Wang, and X. Zuo, "MFEFNet: A multi-scale feature information extraction and fusion network for multi-scale object detection in UAV aerial images," *Drones*, vol. 8, no. 5, p. 186, May 2024.

[21] H. Li and H. Qu, "DASSF: Dynamic-attention scale-sequence fusion for aerial object detection," in *Proc. Int. Conf. Comput. Vis. Media*. Cham, Switzerland: Springer, 2025, pp. 212–227.

[22] H. Li and H. Qu, "VMC-net: multi-scale feature aggregation and distribution with contextual attention guided fusion for aerial object detection," *Complex Intell. Syst.*, vol. 11, no. 8, p. 350, Aug. 2025.

[23] J. Liu et al., "Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Apr. 2022, pp. 5802–5811.

[24] (2019). *Free Flir Thermal Dataset for Algorithm Training*. [Online]. Available: https://www.flir.com/oem/adas/adas-dataset-form

[25] S. Hwang, J. Park, N. Kim, Y. Choi, and I. S. Kweon, "Multispectral pedestrian detection: Benchmark dataset and baseline," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1037–1045.

[26] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou, "LLVIP: A visible-infrared paired dataset for low-light vision," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 3496–3504.

[27] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," in *Proc. J. Vis. Commun. Image Represent.*, vol. 34, 2016, pp. 187–203.

[28] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6700–6713, Oct. 2022.

[29] H. Tang, Z. Li, D. Zhang, S. He, and J. Tang, "Divide-and-conquer: Confluent triple-flow network for RGB-T salient object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1958–1974, Mar. 2025.

[30] H. Zhou et al., "Deformation-resilient multigranularity learning for unaligned RGB-T semantic segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 18530–18544, Oct. 2025.

[31] Z. Chen, W. Xiang, Z. Lin, K. Yang, Y. Liu, and Z. Shi, "Alignment-assisted frequency fusion network for RGB-infrared vehicle detection," *Neurocomputing*, vol. 647, Sep. 2025, Art. no. 130505.

[32] Z. Zheng et al., "GAANet: Graph aggregation alignment feature fusion for multispectral object detection," *IEEE Trans. Ind. Informat.*, vol. 21, no. 11, pp. 8282–8292, Nov. 2025.

[33] H. Zhong, F. Tang, Z. Chen, H. J. Chang, and Y. Gao, "AMDANet: Attention-driven multi-perspective discrepancy alignment for RGB-infrared image fusion and segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 10645–10655.

[34] H. Li, Z. Yang, Y. Zhang, W. Jia, Z. Yu, and Y. Liu, "MulFS-CAP: Multimodal fusion-supervised cross-modality alignment perception for unregistered infrared-visible image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 5, pp. 3673–3690, May 2025.

[35] J. Wang et al., "CIFDet: Robust correlation-guided fusion and contrast-driven attention for misaligned multi-modal object detection," *Inf. Fusion*, vol. 127, Mar. 2026, Art. no. 103833.

[36] Z. Xiong, Z. Yao, X. Liu, W. Zhao, J. Cao, and X. Wu, "Efficient multi-spectral object detection with attentive feature aggregation leveraging zero-shot implicit illumination guidance," *Inf. Fusion*, vol. 118, Jun. 2025, Art. no. 102939.

[37] S. Ding, J. Huang, Z. Cai, Y. Ma, K. Wu, and F. Fan, "FIAFusion: A feedback-based illumination-adaptive infrared and visible image fusion method," *IEEE Sensors J.*, vol. 25, no. 4, pp. 7667–7680, Feb. 2025.

[38] S. Xu, H. Li, T. Liu, and H. Gao, "A method for airborne small-target detection with a multimodal fusion framework integrating photometric perception and cross-attention mechanisms," *Remote Sens.*, vol. 17, no. 7, p. 1118, Mar. 2025.

[39] J. Xu, B. Mo, J. Zhao, C. Zhao, Y. Tao, and S. Han, "Cross-modal adaptive fusion object detection based on illumination-awareness," in *Proc. 39th Youth Academic Annu. Conf. Chin. Assoc. Autom. (YAC)*, Jun. 2024, pp. 931–938.

[40] Z. Liu et al., "High-precision pest management based on multimodal fusion and attention-guided lightweight networks," *Insects*, vol. 16, no. 8, p. 850, Aug. 2025.

[41] H. Tang, S. He, and J. Qin, "Connecting giants: Synergistic knowledge transfer of large multimodal models for few-shot learning," 2025, *arXiv:2510.11115*.

[42] H. Tang, Y. Liu, S. Yan, F. Shen, S. He, and J. Qin, "Cross-modal proxy evolving for OOD detection with vision-language models," 2026, *arXiv:2601.08476*.


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[43] A. Dosovitskiy et al., "An image is worth $16 \times 16$ words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[44] M. M. Adnan et al., "Image annotation with YCbCr color features based on multiple deep CNN- GLP," *IEEE Access*, vol. 12, pp. 11340–11353, 2024.
[45] J. Zhang, J. Wu, Y. Chen, Y. Deng, J. Yang, and X. Li, "An improved green vegetation segmentation method based on RGB and NIR image pairs," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Aug. 2025, pp. 4402–4405.
[46] I. R. Khan, S. Rahardja, M. M. Khan, M. M. Movania, and F. Abed, "A tone-mapping technique based on histogram using a sensitivity model of the human visual system," *IEEE Trans. Ind. Electron.*, vol. 65, no. 4, pp. 3469–3479, Apr. 2018.
[47] Y. Zhang et al., "Drone-based RGBT tiny person detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 61–76, Oct. 2023.
[48] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2844–2853.
[49] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.
[50] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented reppoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1829–1838.
[51] G. Jocher, J. Qiu, and A. Chaurasia. (2023). *Ultralytics YOLO*. [Online]. Available: https://github.com/ultralytics/ultralytics
[52] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.
[53] M. Sharma et al., "YOLOrs: Object detection in multimodal remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1497–1508, 2021.
[54] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1695–1708, 2023.
[55] J. Zhang, J. Lei, W. Xie, Y. Li, G. Yang, and X. Jia, "Guided hybrid quantization for object detection in remote sensing imagery via one-to-one self-teaching," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023.
[56] Y. Zhu, X. Sun, M. Wang, and H. Huang, "Multi-modal feature pyramid transformer for RGB-infrared object detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 9, pp. 9984–9995, Sep. 2023.
[57] J. Shen, Y. Chen, Y. Liu, X. Zuo, H. Fan, and W. Yang, "ICAFusion: Iterative cross-attention guided feature fusion for multispectral object detection," *Pattern Recognit.*, vol. 145, Jan. 2024, Art. no. 109913.
[58] Y. Xiao, F. Meng, Q. Wu, L. Xu, M. He, and H. Li, "GM-DETR: Generalized muiltispectral detection transformer with efficient fusion encoder for visible-infrared detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 5541–5549.
[59] A. Althoupety, L.-Y. Wang, W.-C. Feng, and B. Rekabdar, "DaFF: Dual attentive feature fusion for multispectral pedestrian detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 2997–3006.
[60] K. Song, X. Xue, H. Wen, Y. Ji, Y. Yan, and Q. Meng, "Misaligned visible-thermal object detection: A drone-based benchmark and baseline," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 11, pp. 7449–7460, Nov. 2024.
[61] G. Jocher, J. Qiu, and A. Chaurasia, "Ultralytics YOLO," 2024. [Online]. Available: https://github.com/ultralytics/ultralytics
[62] C. Liu, X. Ma, X. Yang, Y. Zhang, and Y. Dong, "COMO: Cross-Mamba interaction and offset-guided fusion for multimodal object detection," *Inf. Fusion*, vol. 125, Jan. 2026, Art. no. 103414.
[63] P. Sun et al., "Sparse R-CNN: End-to-end object detection with learnable proposals," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14454–14463.
[64] H. Zhang et al., "DINO: DETR with improved denoising anchor boxes for end-to-end object detection," in *Proc. ICLR*, 2023.
[65] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.
[66] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: A simple and strong anchor-free object detector," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 1922–1933, Apr. 2022.
[67] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.
[68] X. Li et al., "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21002–21012.
[69] F. Qingyun and W. Zhaokui, "Cross-modality attentive feature fusion for object detection in multispectral remote sensing imagery," *Pattern Recognit.*, vol. 130, Oct. 2022, Art. no. 108786.
[70] Y. T. Chen, J. Shi, Z. Ye, C. Mertz, D. Ramanan, and S. Kong, "Multimodal object detection via probabilistic ensembling," in *Proc. Eur. Conf. Comput. Vis*. Switzerland: Springer, Oct. 2022, pp. 139–158.
[71] T. Brödermann, C. Sakaridis, D. Dai, and L. Van Gool, "HRFuser: A multi-resolution sensor fusion architecture for 2D object detection," in *Proc. IEEE 26th Int. Conf. Intell. Transp. Syst. (ITSC)*, Sep. 2023, pp. 4159–4166.
[72] P. Peng, T. Xu, L. Song, M. Zhu, Y. Fang, and J. Li, "COXNet: Cross-layer fusion with adaptive alignment and scale integration for RGBT tiny object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 1, pp. 596–608, Jan. 2026.
[73] T. Zhao, M. Yuan, F. Jiang, N. Wang, and X. Wei, "Removal then selection: A coarse-to-fine fusion perspective for RGB-infrared object detection," 2024, *arXiv:2401.10731*.
[74] M. Hong, S. Li, Y. Yang, F. Zhu, Q. Zhao, and L. Lu, "SSPNet: Scale selection pyramid network for tiny person detection from UAV images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[75] C. Xu, J. Wang, W. Yang, and L. Yu, "Dot distance for tiny object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1192–1201.
[76] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 79–93, Aug. 2022.
[77] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.
[78] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13658–13667.
[79] X. Zhang et al., "TFDet: Target-aware fusion for RGB-T pedestrian detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 13276–13290, Jul. 2025.
[80] Y. Tang, X. Tong, Z. He, J. Yang, and Y. Cao, "Adaptive modality selection drone-based RGBT detector for tiny targets," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–14, 2025.
[81] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Feb. 2017, pp. 764–773.
[82] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[83] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "Cbam: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, May 2018, pp. 3–19.
[84] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[85] Z. Chen, C. Yang, Q. Li, F. Zhao, Z.-J. Zha, and F. Wu, "Disentangle your dense object detector," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 4939–4948.



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